

A Computational Theory of Vowel System Typology Based on Optimal Listeners and Speakers



John McGahay

University of California, Los Angeles

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Acknowledgements

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 - $\sum_{i=1}^{n-1} \sum_{j=i+1}^n 1/d_{ij}^2$ where d_{ij} is pairwise psychoacoustic distance between vowels i and j
- Optimal systems with minimal energy resemble typologically most common systems
 - e.g. /iau/, /ieau/, /ieaou/, etc.

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Lack of mechanism for system emergence

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- **Iterated Confusion Minimization (ICM)**

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Such accounts model phonetic categories as probability distributions over phonetic continua

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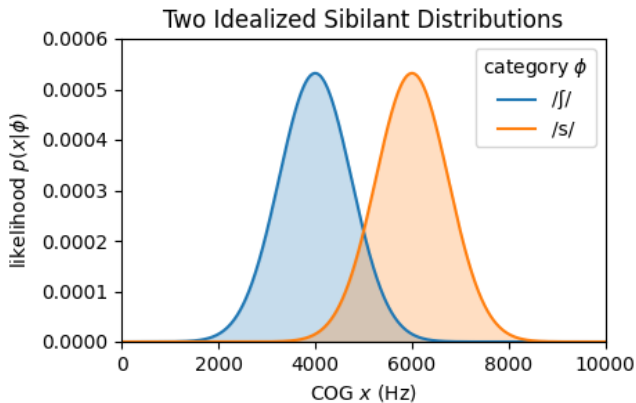
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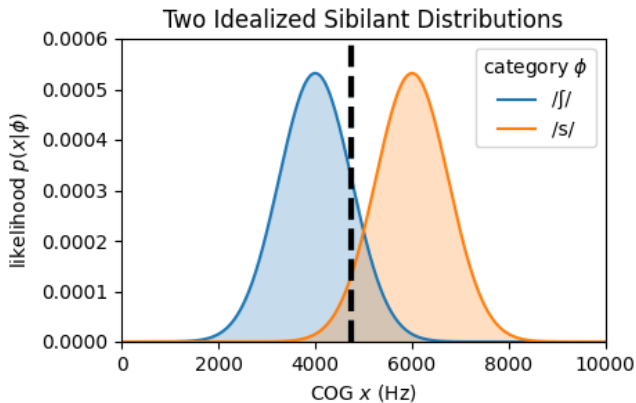
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- allows for mathematical characterization of rational listener behavior

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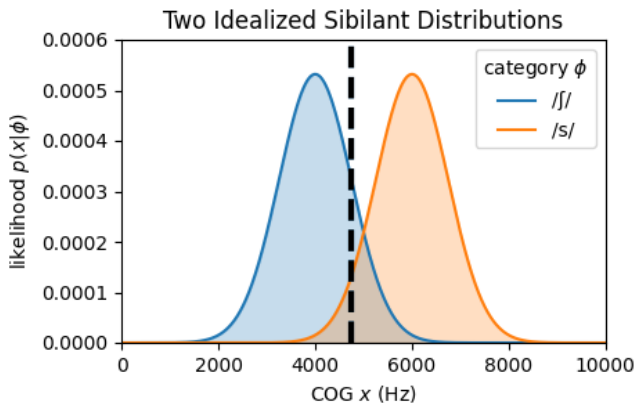


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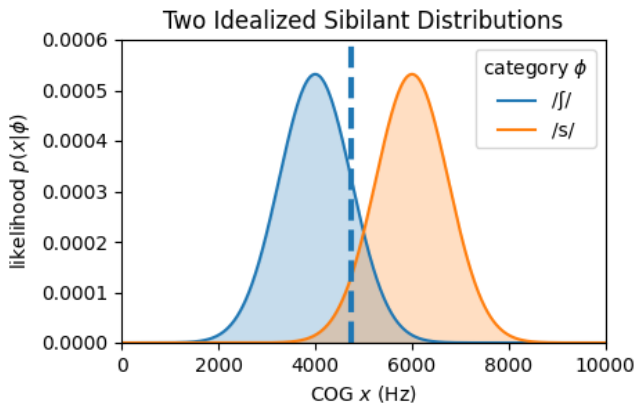
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Classifying phonetic tokens as most likely category maximizes accuracy



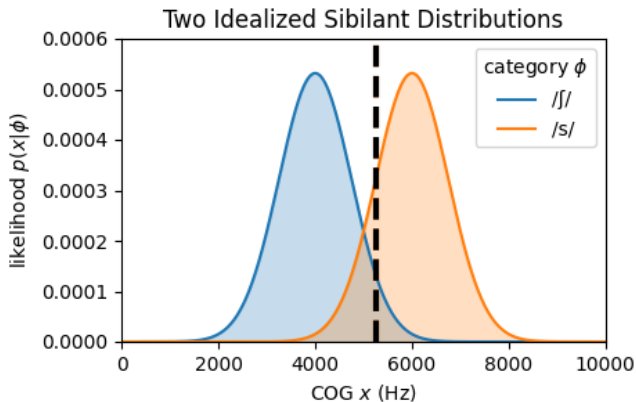
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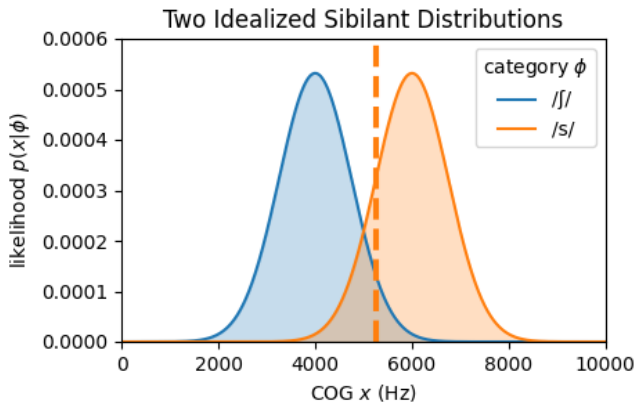
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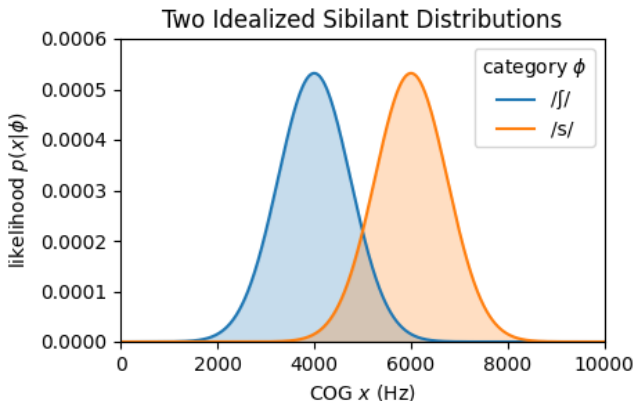
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where:

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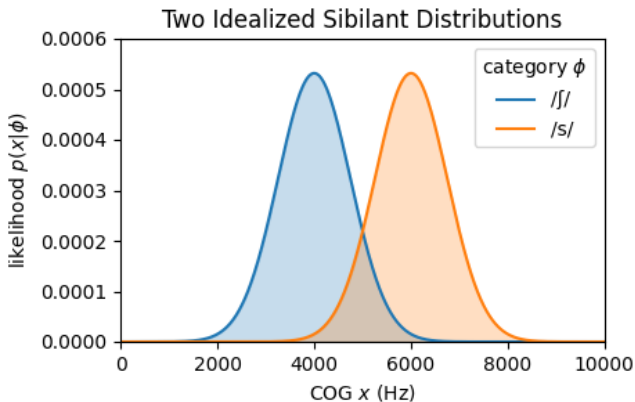
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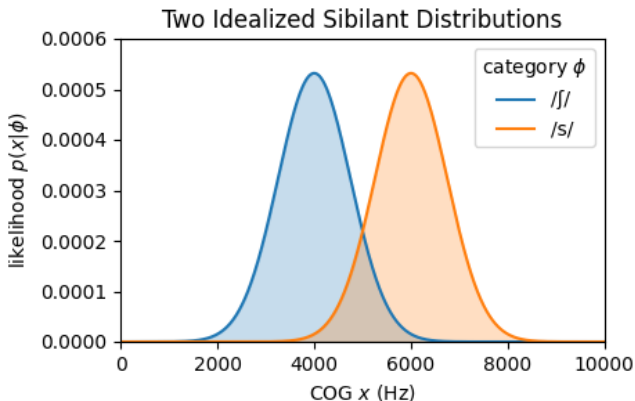
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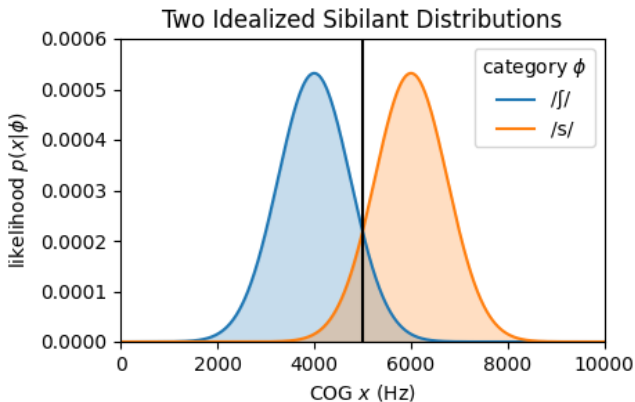
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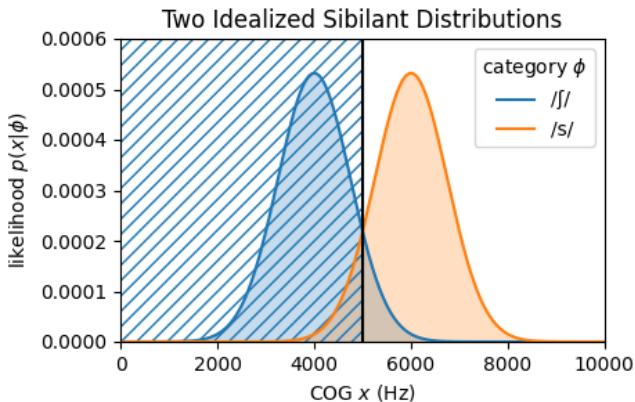
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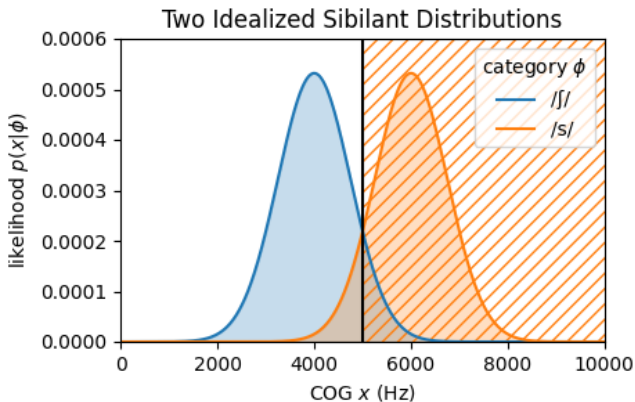
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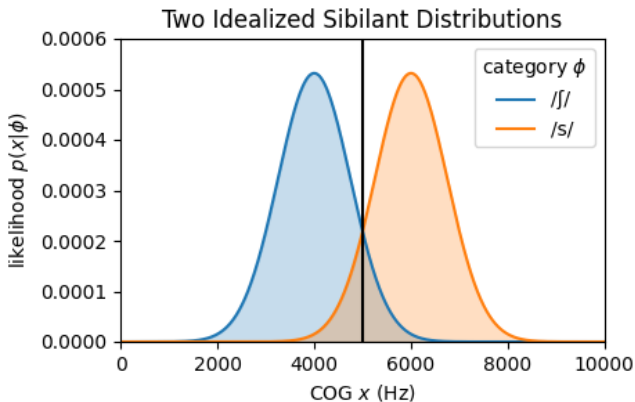
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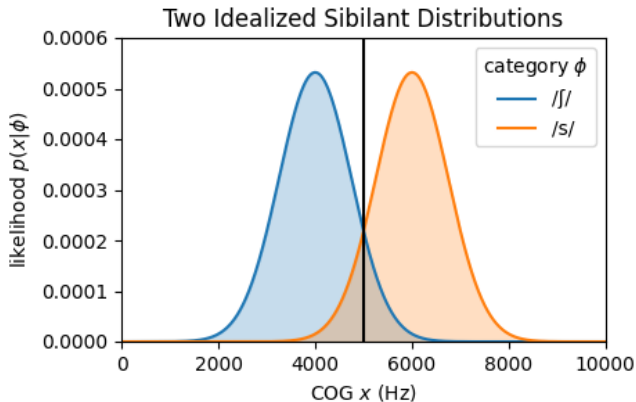
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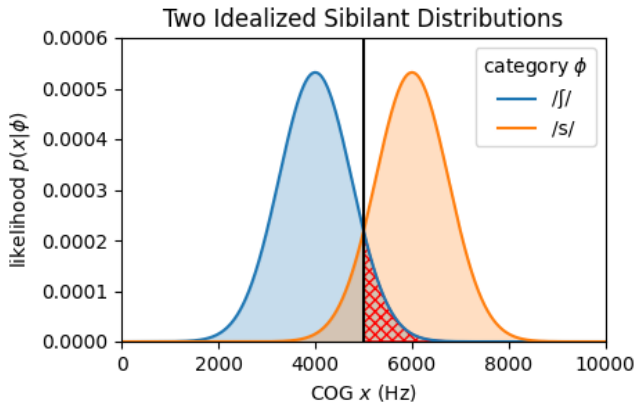
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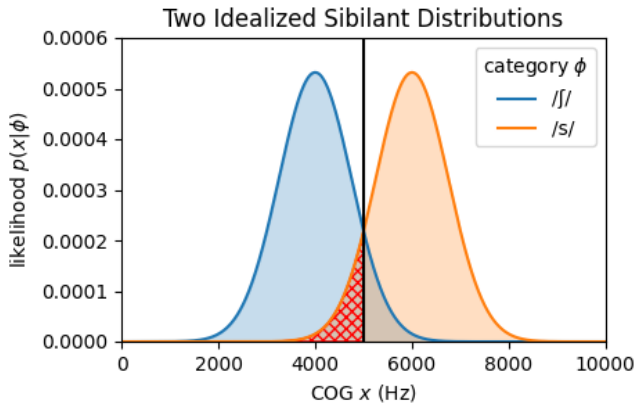
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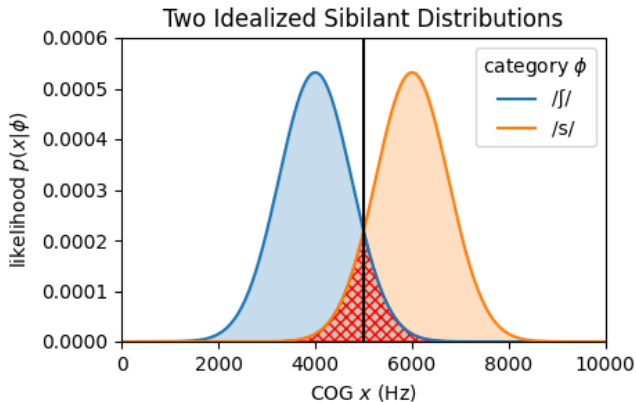
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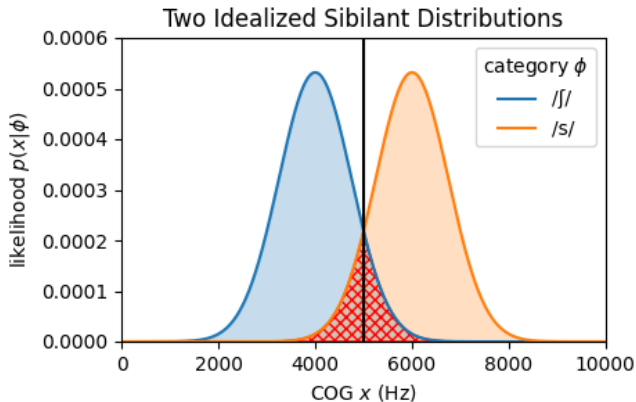
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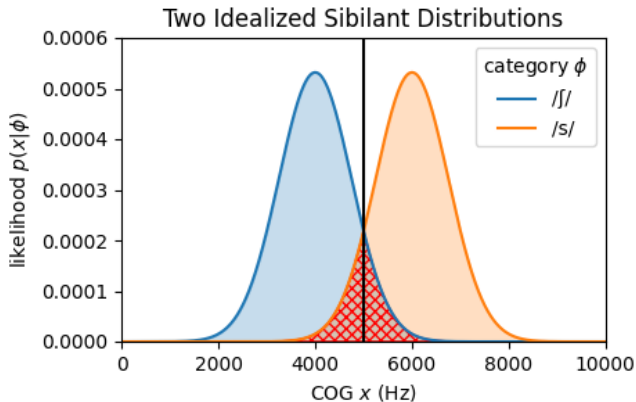
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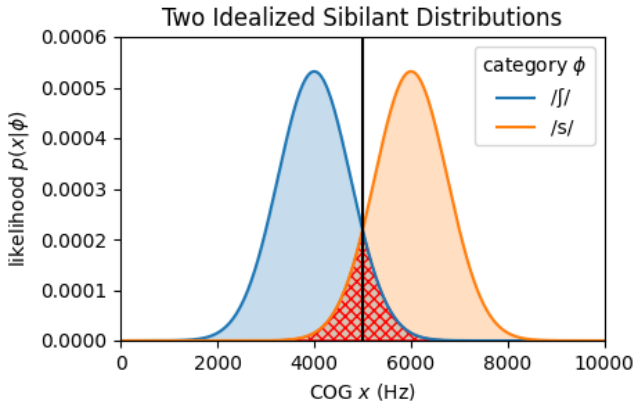
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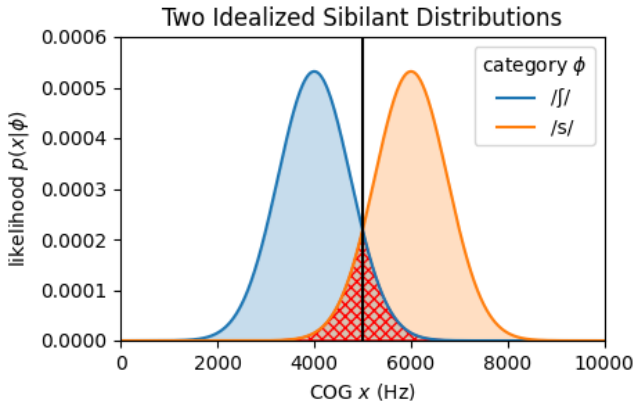
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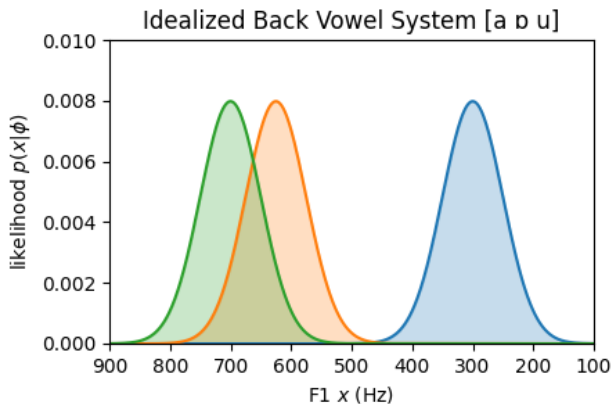
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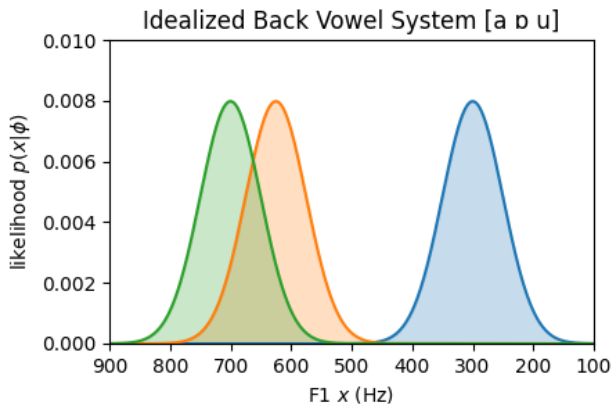
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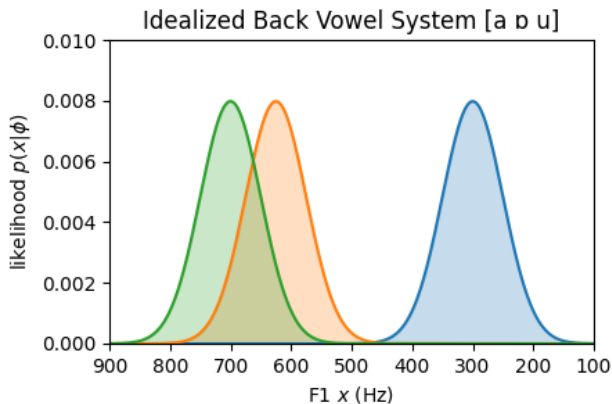


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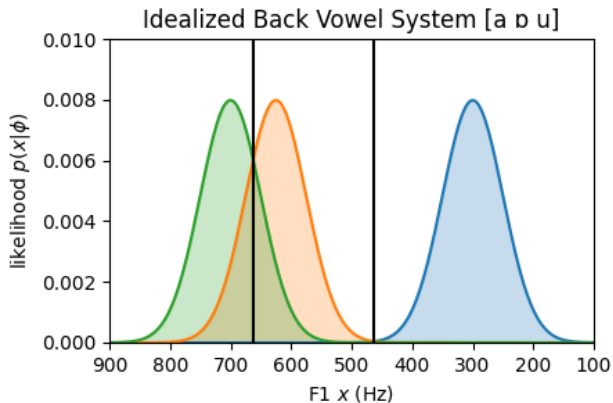


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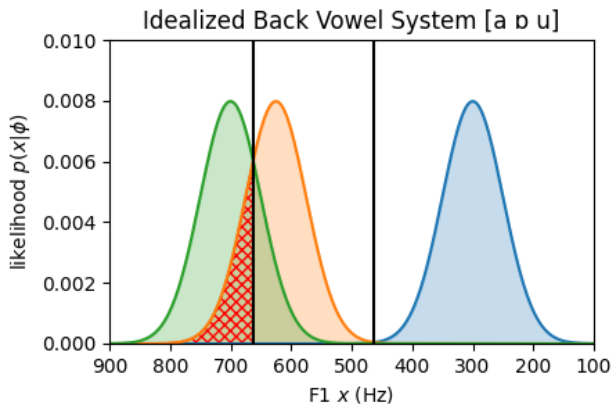


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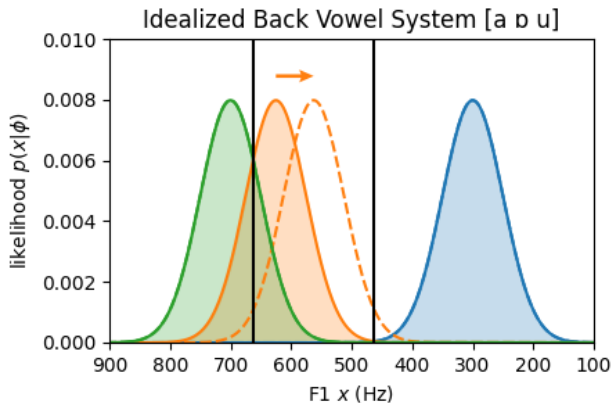


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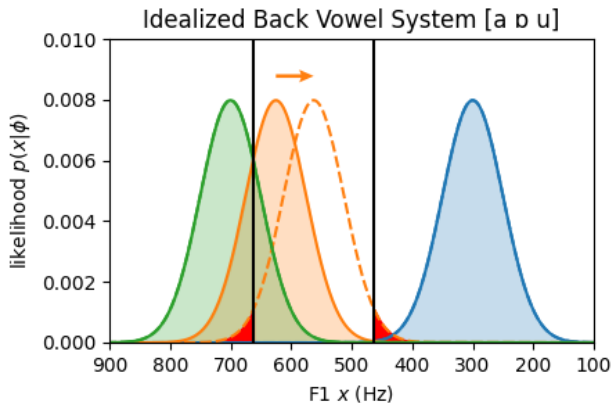


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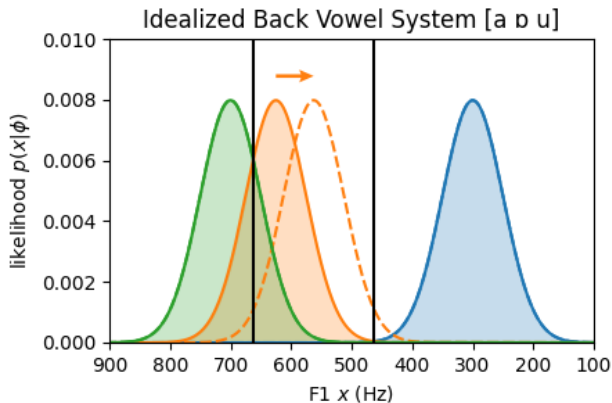
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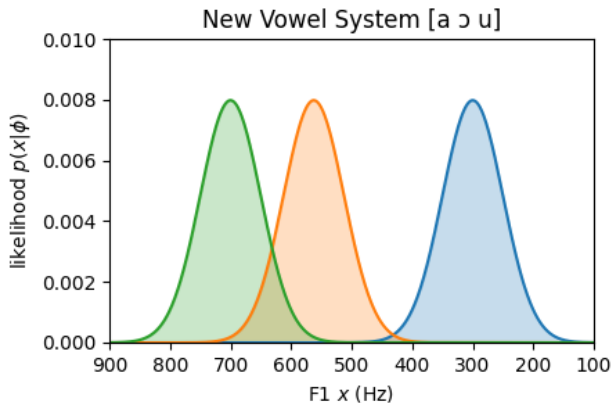
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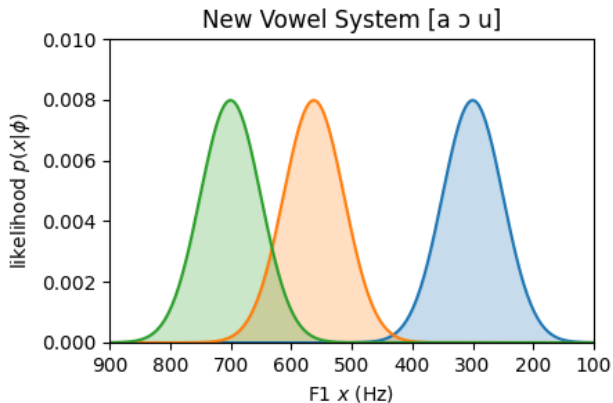
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Right: idealized back vowel system
[a ɔ u] modeled as F1 distributions

- Typologically unusual; [a ɔ u] expected (Becker-Kristal, 2010)

A new set of distributions can reduce confusion, *even if the listener expects the original distributions*

- Centering a distribution between its boundaries reduces confusion
- New system is more dispersed and less typologically unusual



Optimal Speaker Behavior

optimal speaker behavior

A speaker can minimize the confusion of a maximum-likelihood listener by adopting system S' :

$$S' = \left\{ \operatorname{argmin}_{\phi' \in \Phi} \int p(x|\phi') P(\phi \neq C_S(x)) dx \mid \phi \in S \right\}$$

where:

- Φ : set of possible phonetic distributions
- $S \subset \Phi$: pre-existing phonetic system expected by maximum-likelihood listener
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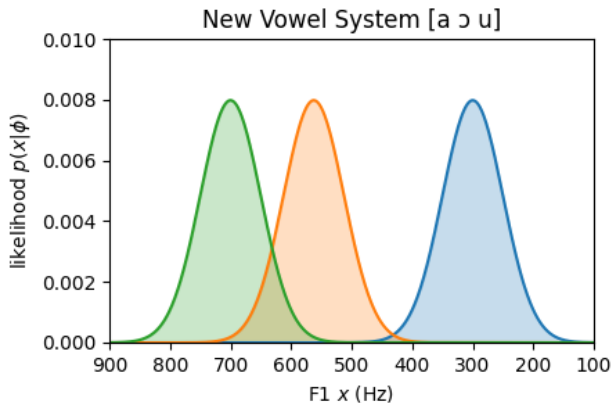
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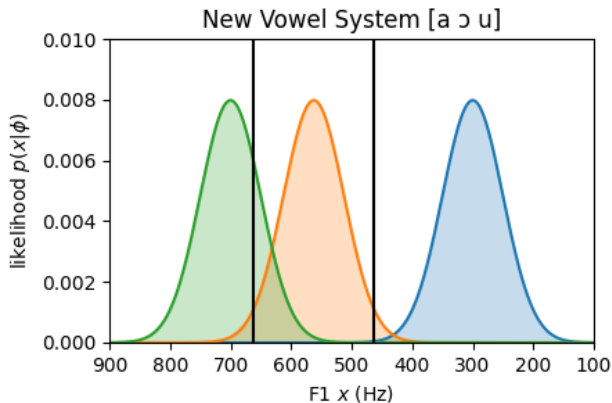
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- e.g. normal distributions $\mathcal{N}(\mu, \sigma^2)$ with F1 mean $300\text{Hz} \leq \mu \leq 700\text{Hz}$ and s.d. $\sigma = 50\text{Hz}$

Iterated Confusion Minimization

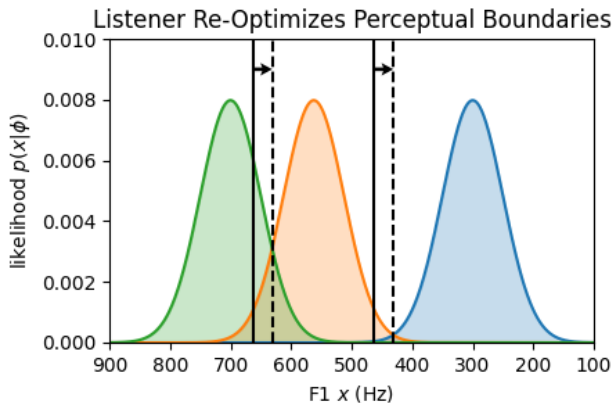


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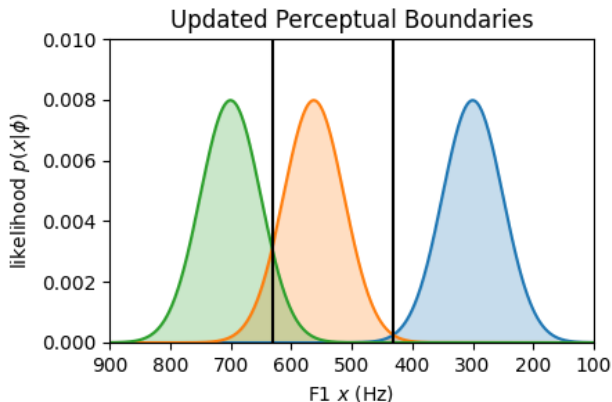
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Given speaker's new system, the optimal listener updates their perceptual boundaries



Iterated Confusion Minimization

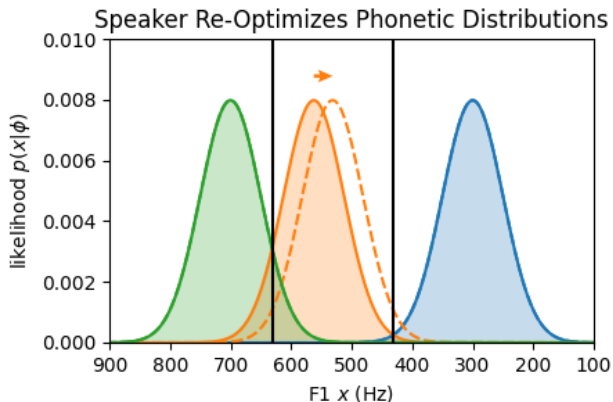
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Given speaker's new system, the optimal listener updates their perceptual boundaries

Then, the optimal speaker updates their phonetic distributions again

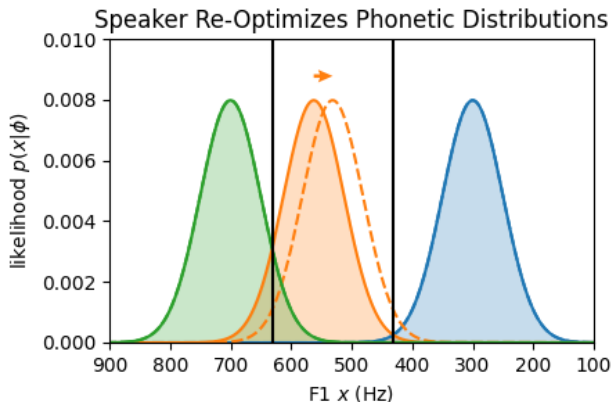


Iterated Confusion Minimization

Given speaker's new system, the optimal listener updates their perceptual boundaries

Then, the optimal speaker updates their phonetic distributions again

We will call this repeated process
Iterated Confusion Minimization (ICM)



Iterated Confusion Minimization Algorithm

Iterated Confusion Minimization (ICM)

Input: $S_0 \subseteq \Phi$, where Φ is set of possible phonetic distributions

Output: $S_i \subseteq \Phi$

1: $i \leftarrow 0$

2: **repeat**

3: ▷ *Optimal listener behavior*

4: $C_i(x) \leftarrow \operatorname{argmax}_{\phi \in S_i} p(x|\phi)$

5: ▷ *Optimal speaker behavior*

6: $S_{i+1} \leftarrow \{\operatorname{argmin}_{\phi' \in \Phi} \int p(x|\phi') P(\phi \neq C_S(x)) dx | \phi \in S_i\}$

7: $i \leftarrow i + 1$

8: **until** $S_i = S_{i-1}$

9: **return** S_i

Implementing ICM over the Vowel Space

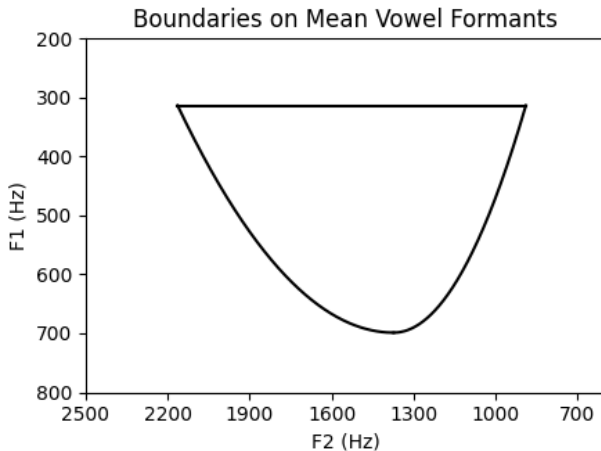
Implementing ICM over the Vowel Space

2D normal distributions over F1 & F2

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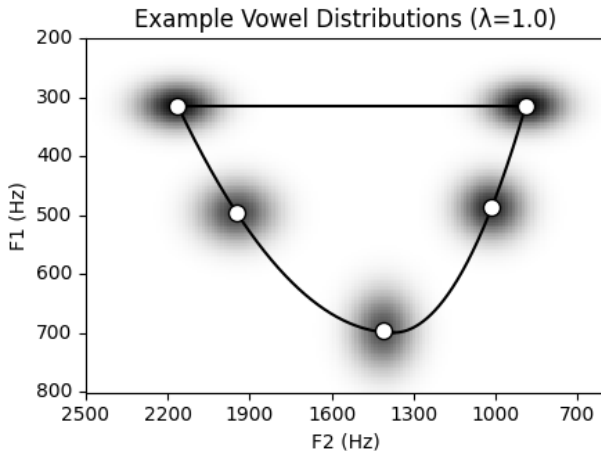
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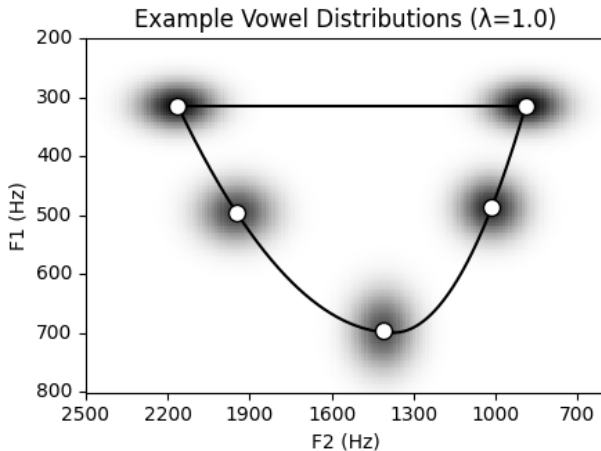
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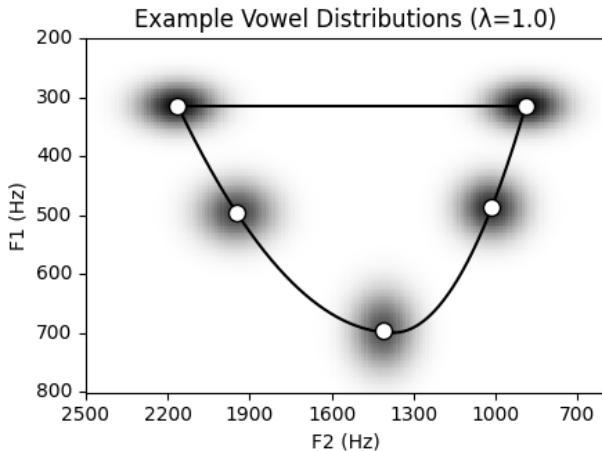
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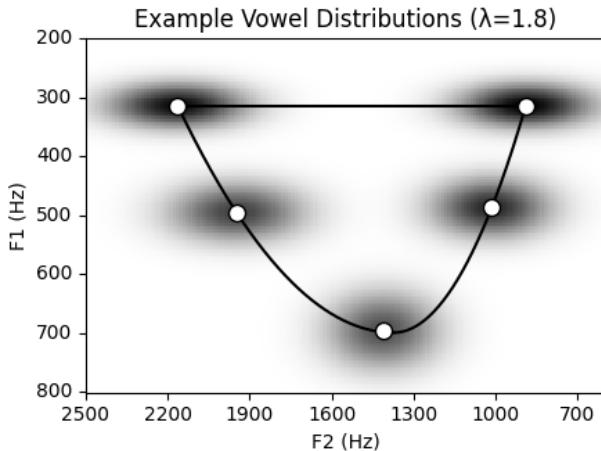
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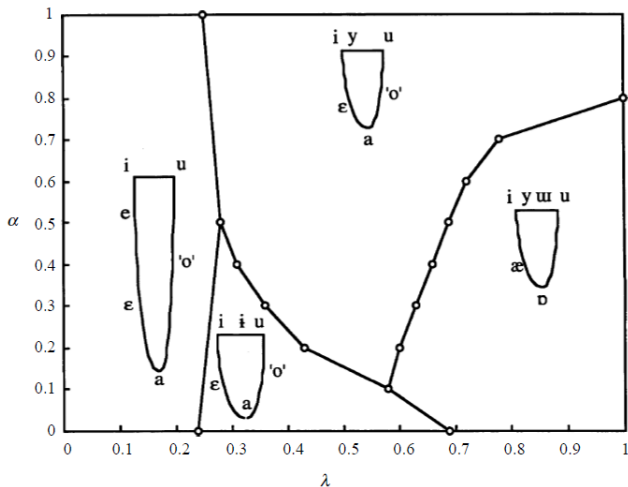
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Schwartz et al.'s (1997b) optimal 6V systems

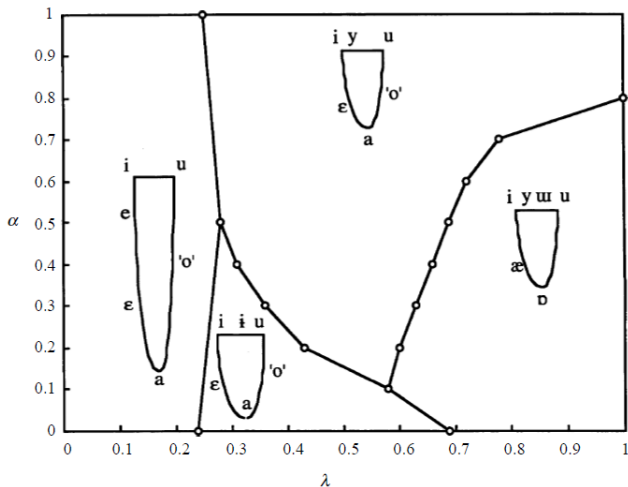


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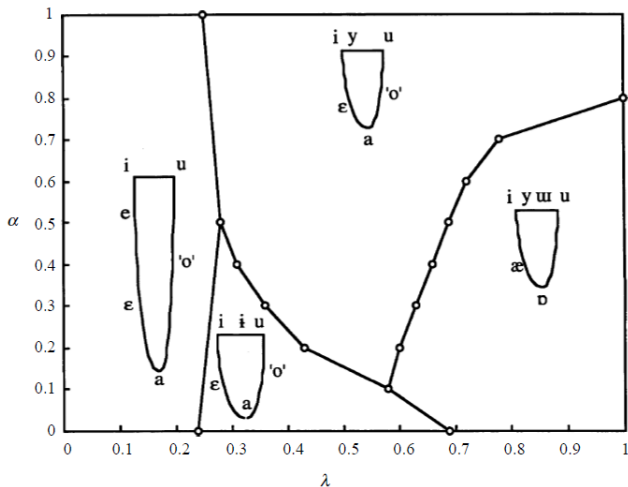


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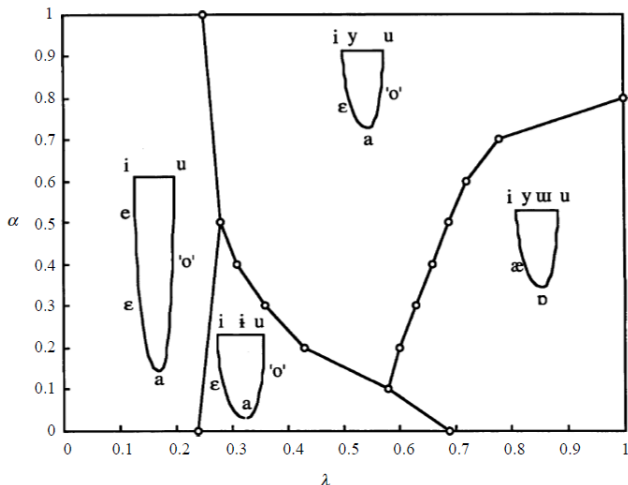
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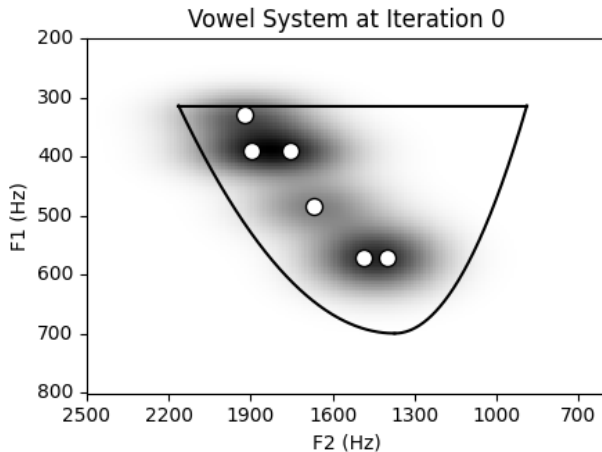
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Scaling up F2 s.d. has analogous effects to scaling down F2 distances in past models

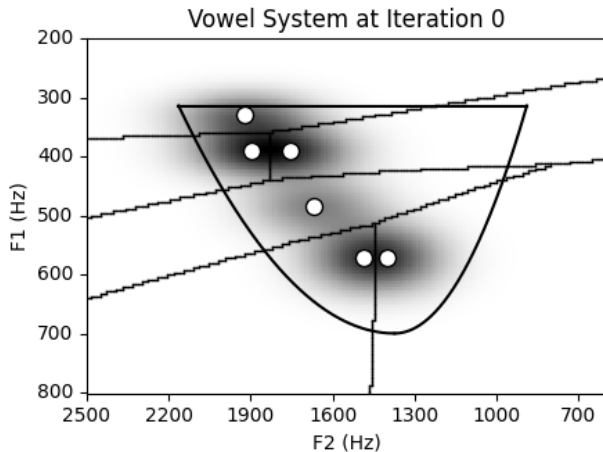
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Example Simulation 1: Chain Shifts

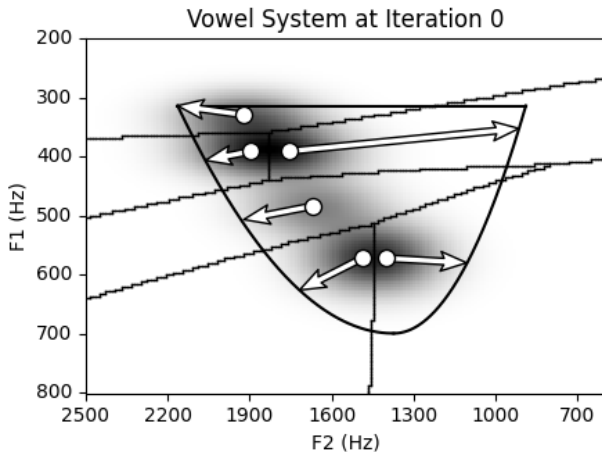


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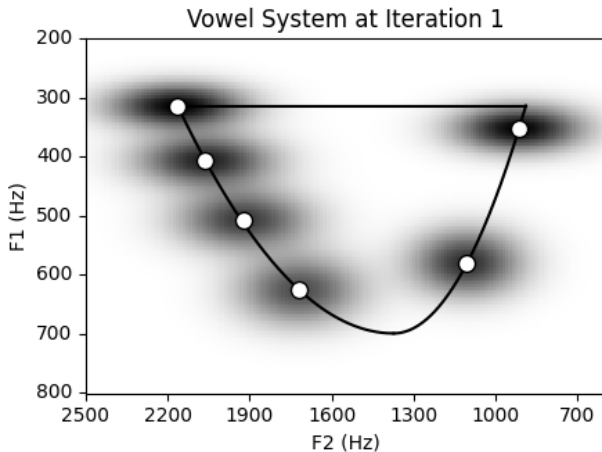
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ICM exerts strong pressure towards vowel space periphery



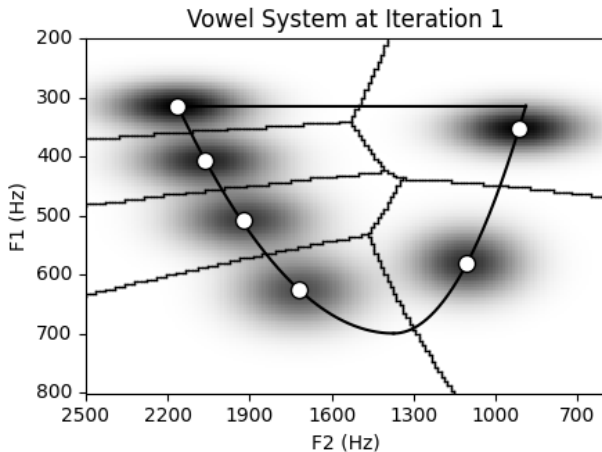
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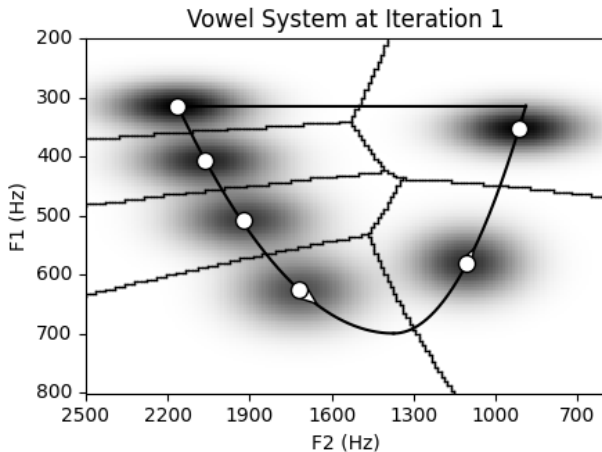
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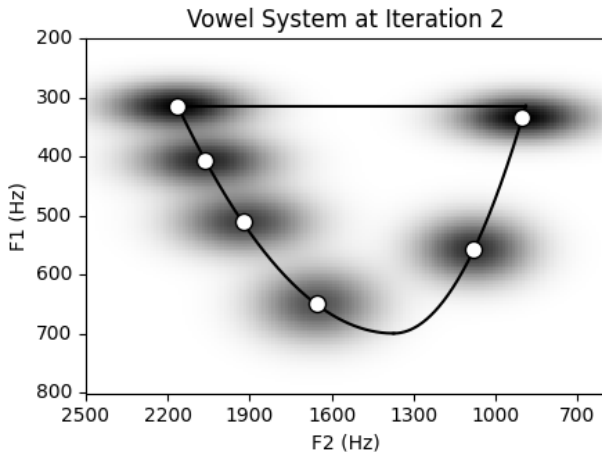
Incremental chain shift reminiscent of real-world sound change



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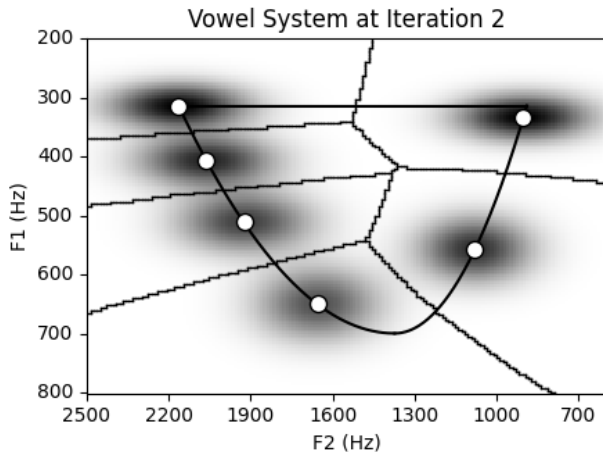
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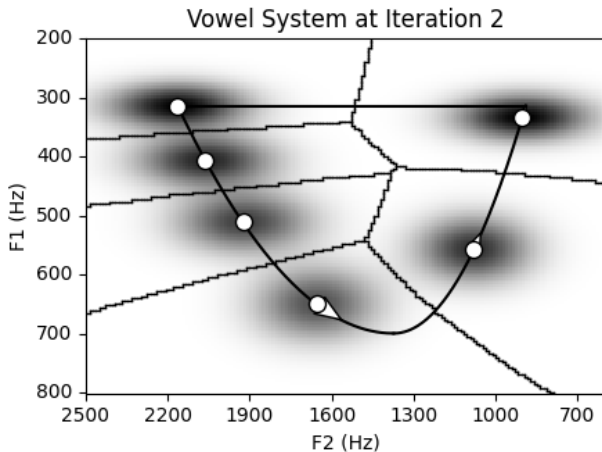
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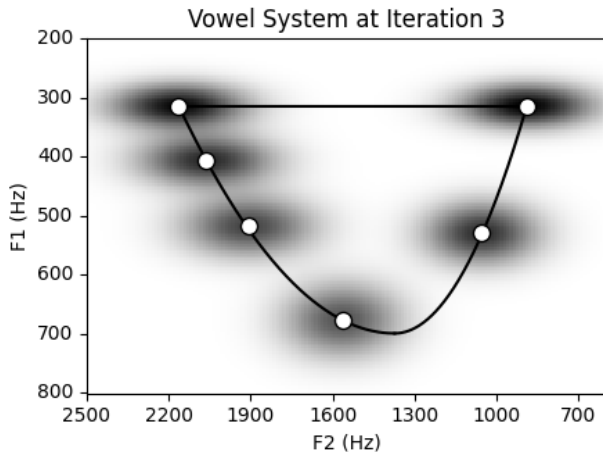
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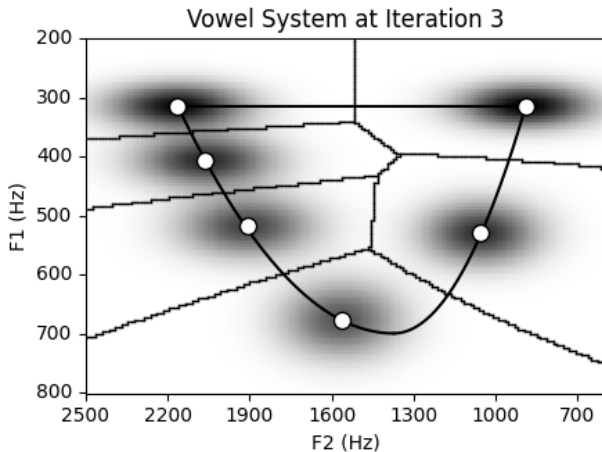
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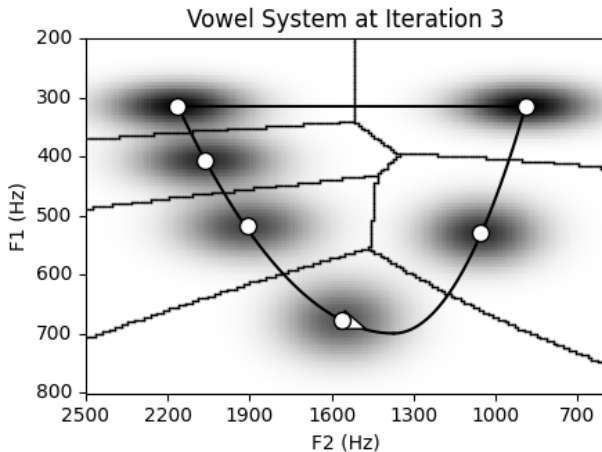
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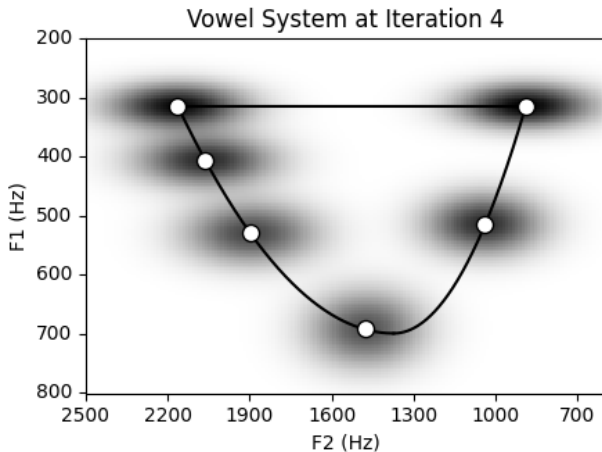
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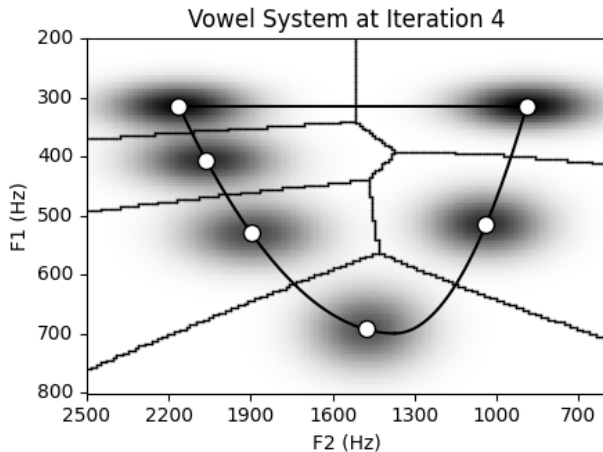
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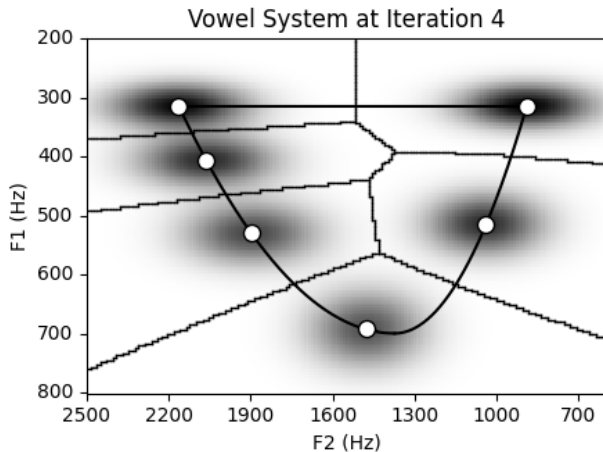
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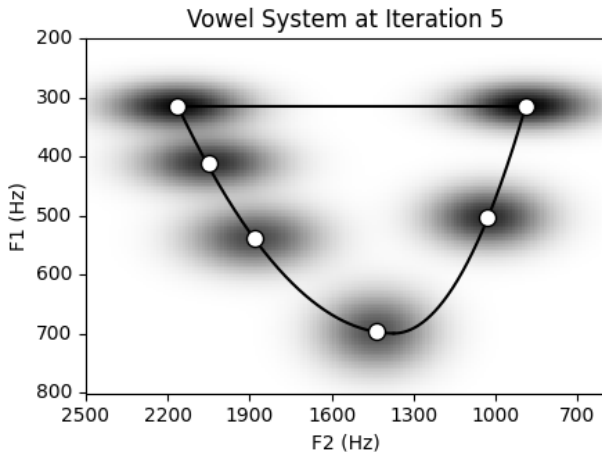
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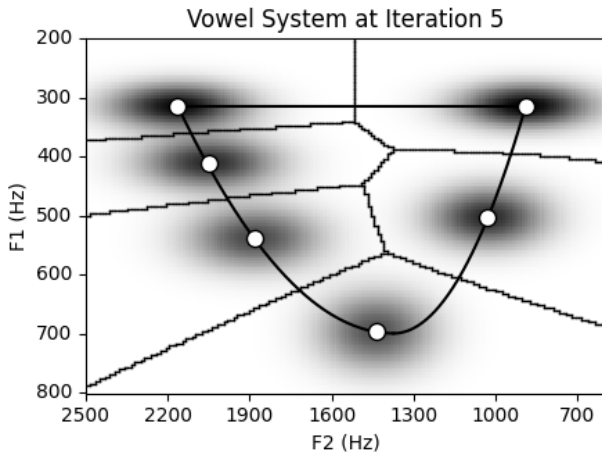
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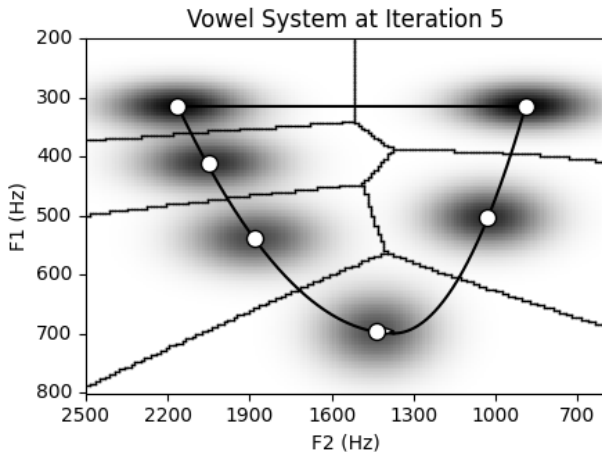
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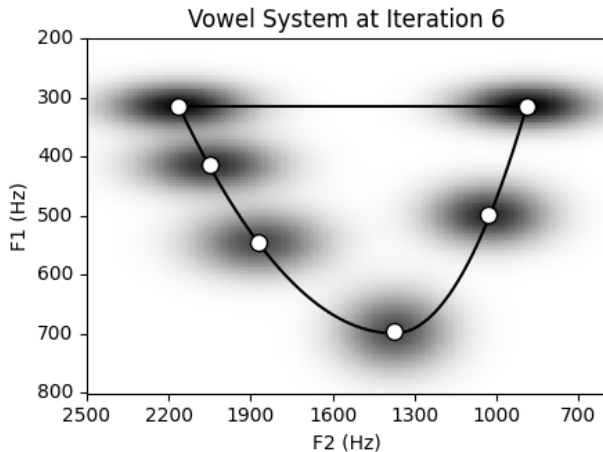
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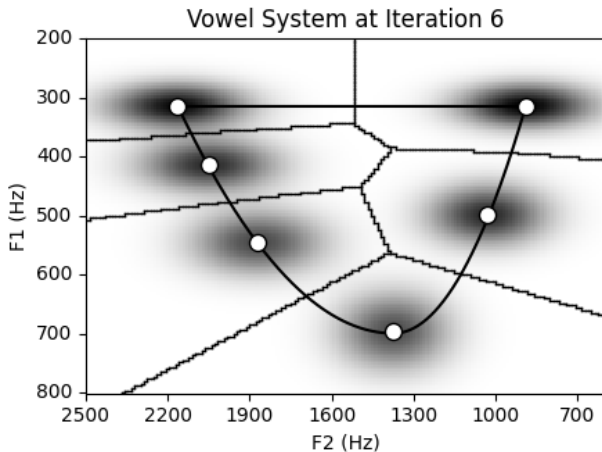
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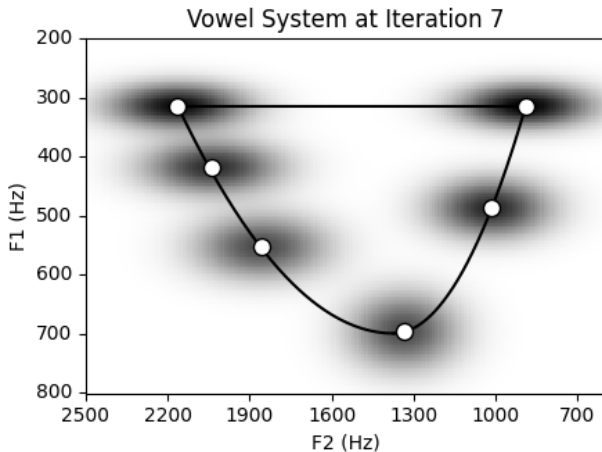
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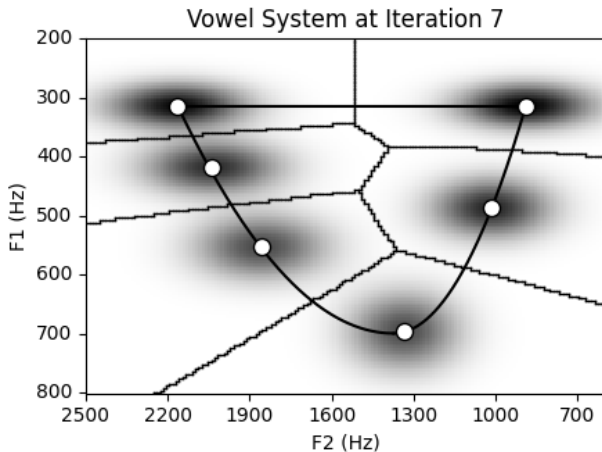
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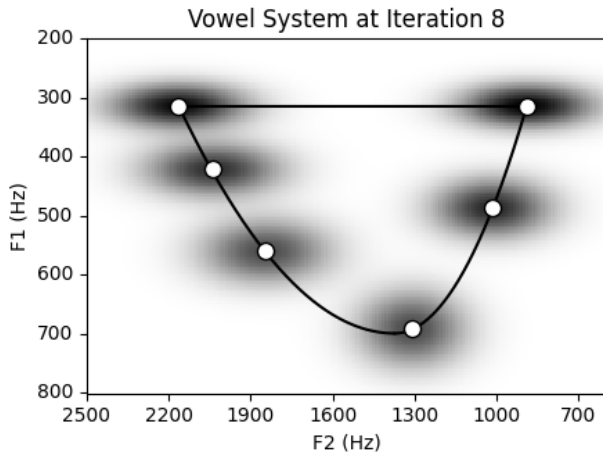
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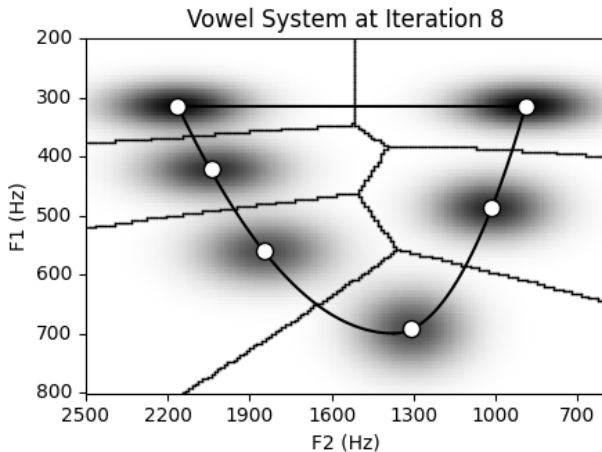
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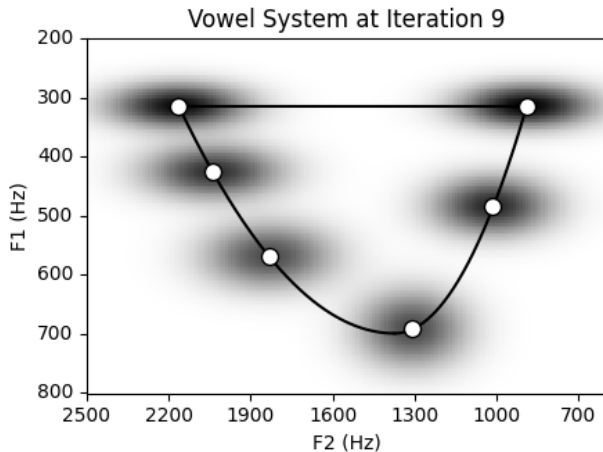
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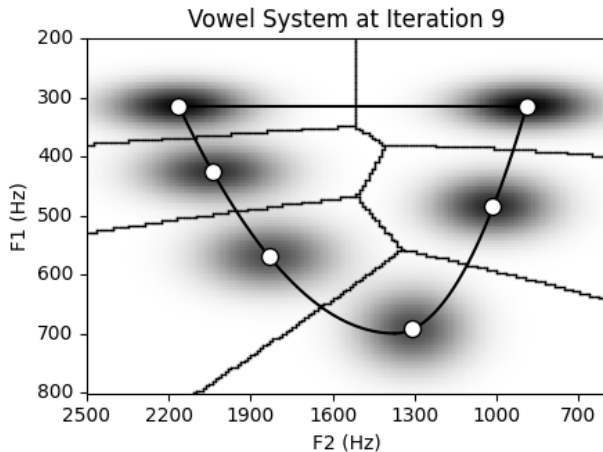
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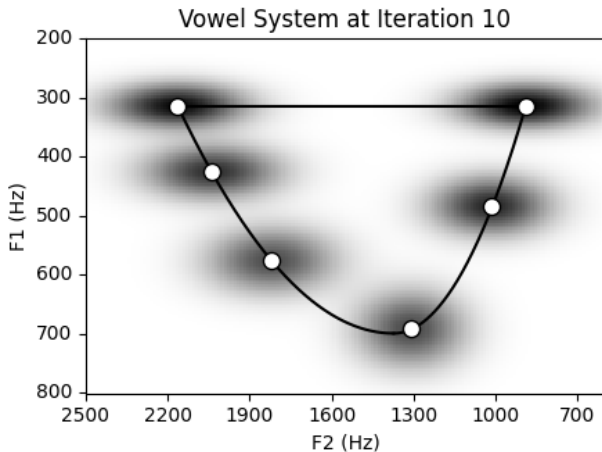
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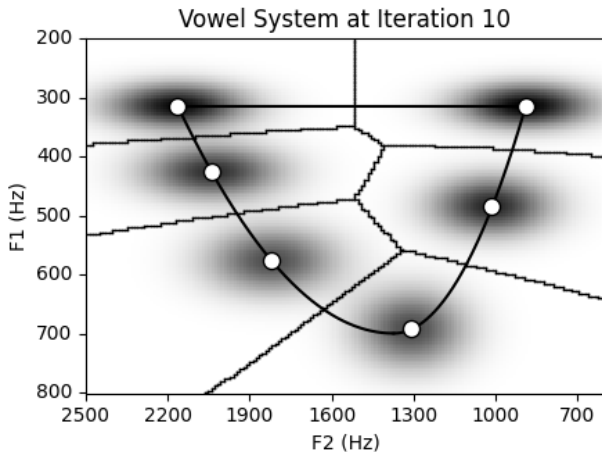
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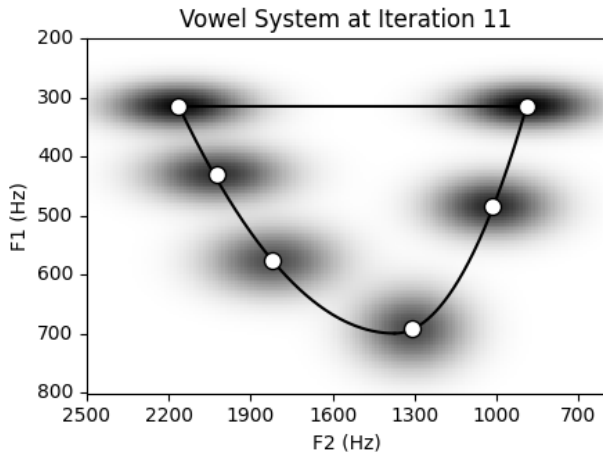
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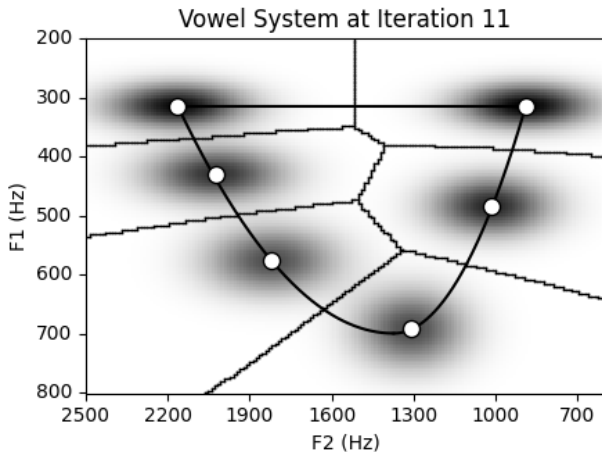
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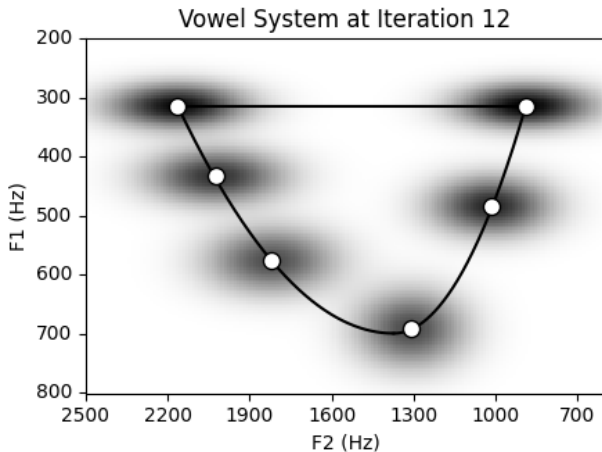
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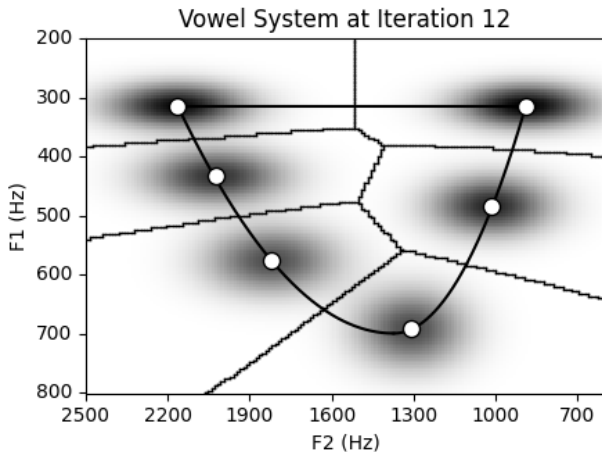


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Incremental chain shift reminiscent of real-world sound change

- cf. California/Canada shift
[i] → [ɛ] → [æ] → [a] → [ɔ]

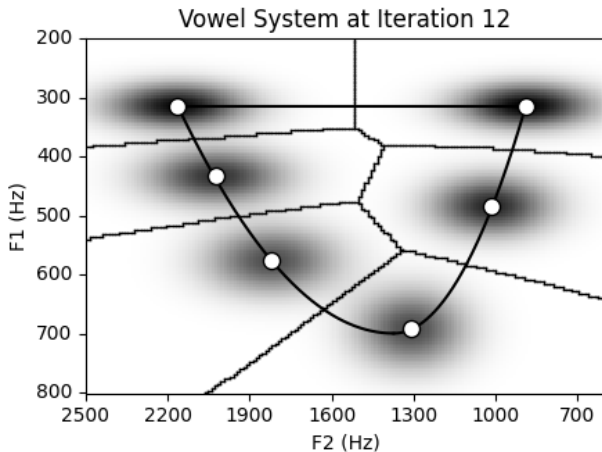


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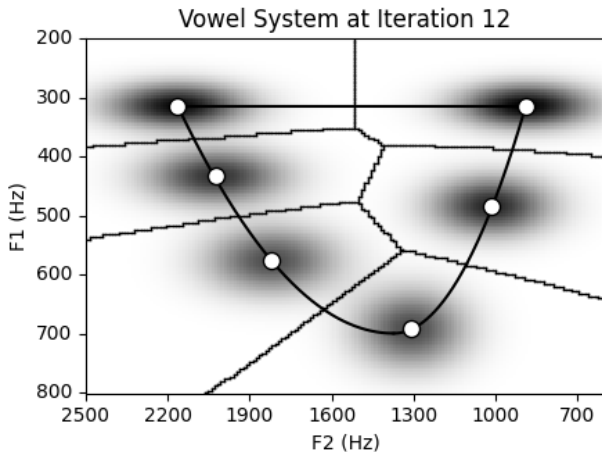
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ICM yields naturalistic equilibrium system with dispersed vowels



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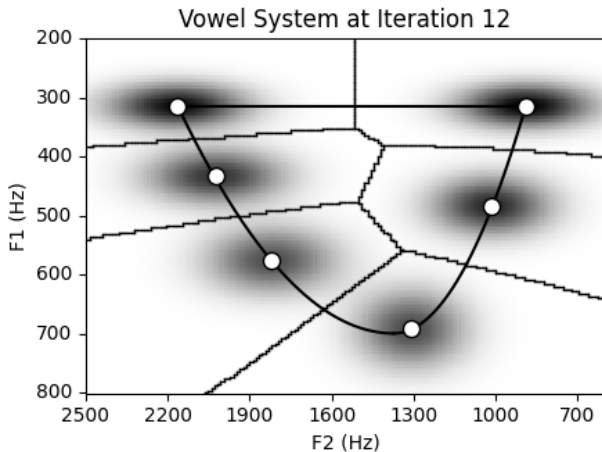
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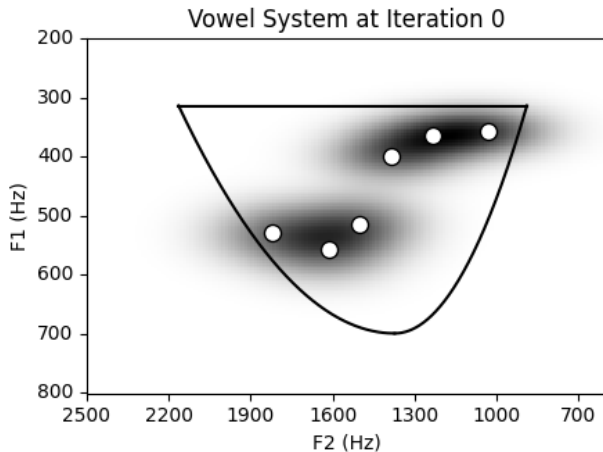
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- /ieɛaou/ 2nd most common 6V system (Schwartz et al., 1997a)



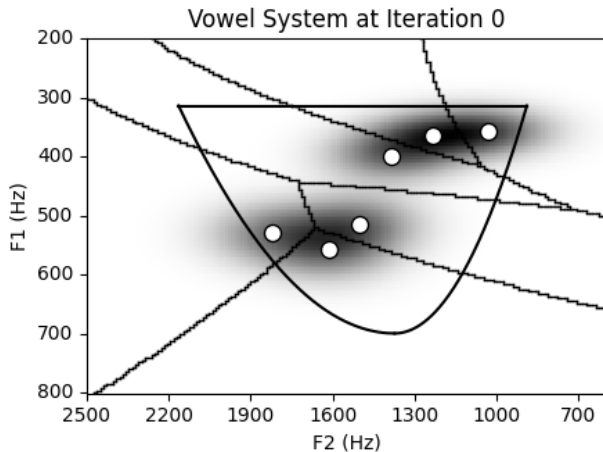
Example Simulation 2: Interior Vowels and Multiple Stable Systems

Most common 6V system /ieaouə/
(Schwartz et al., 1997a) also emerges



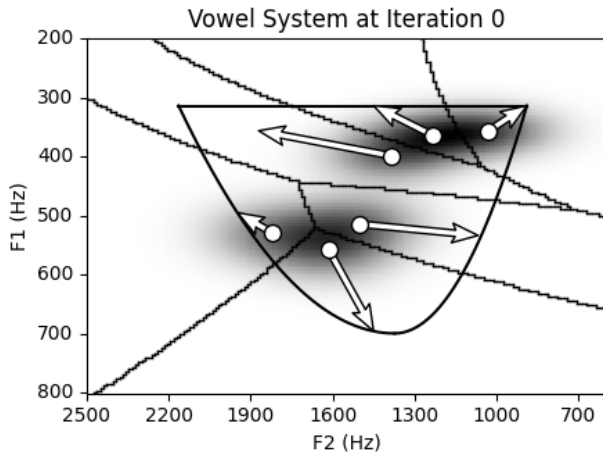
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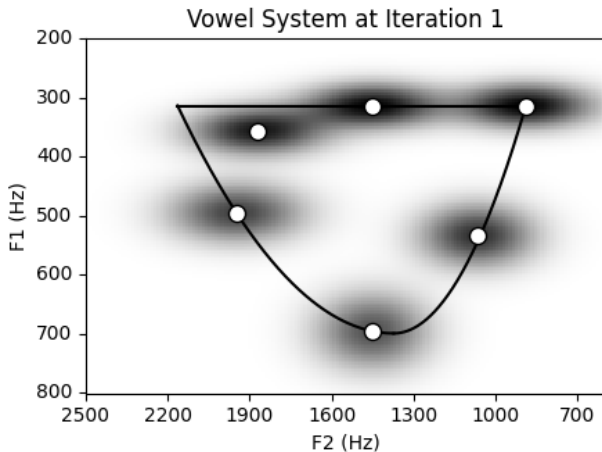
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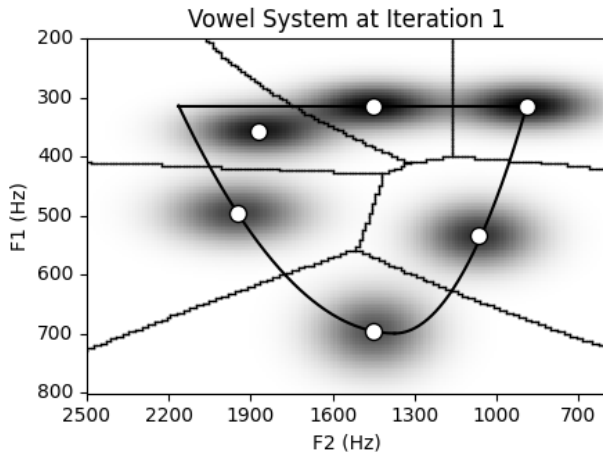
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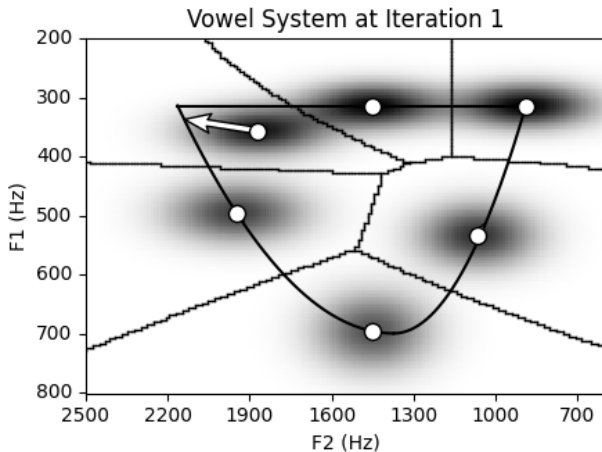
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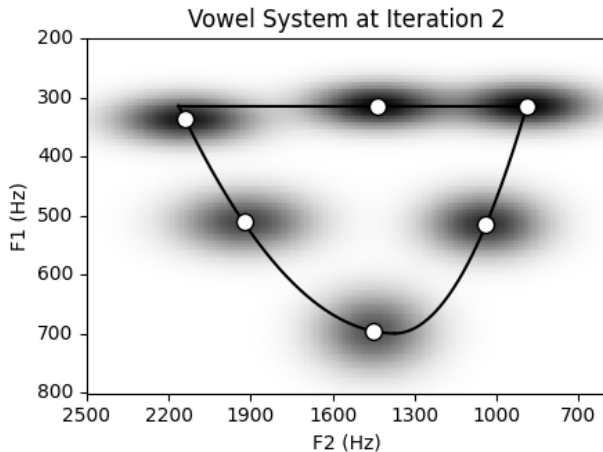
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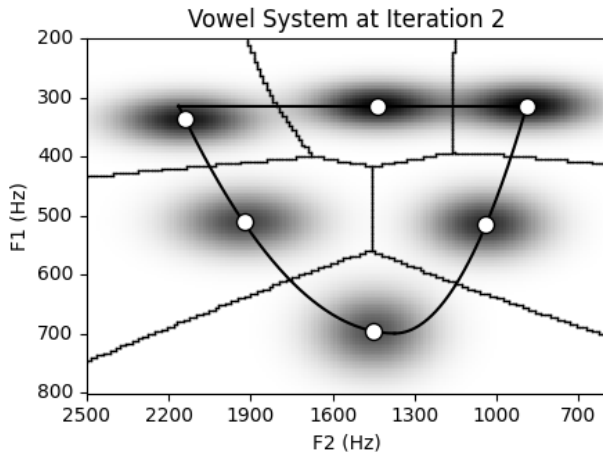
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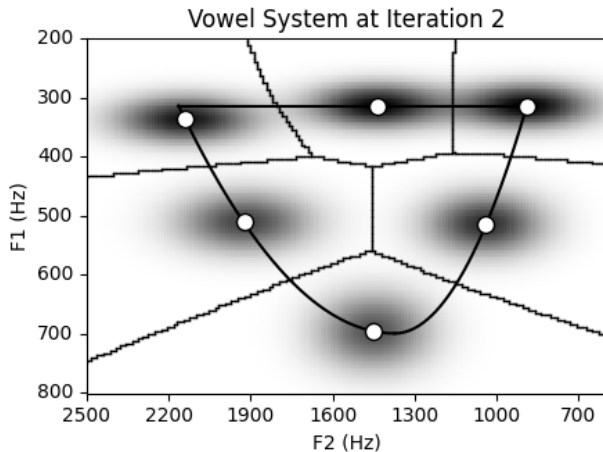
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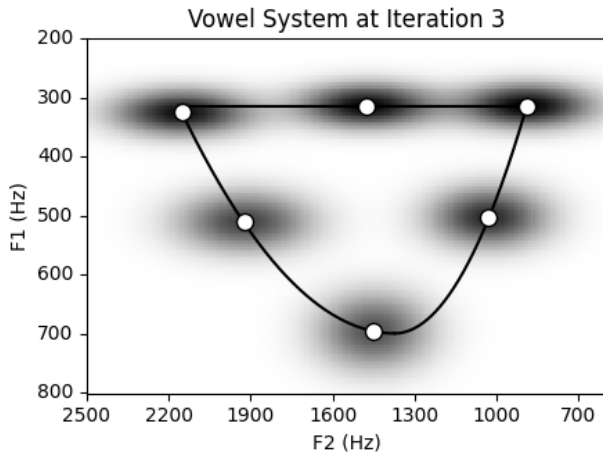
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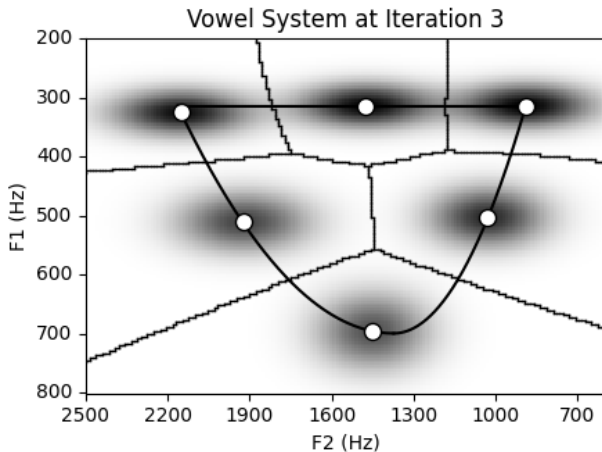
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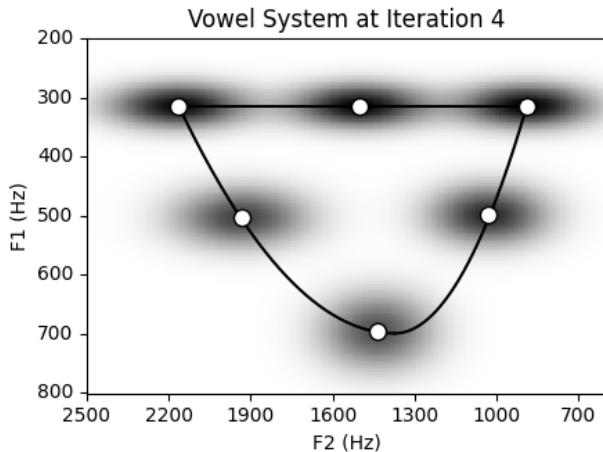
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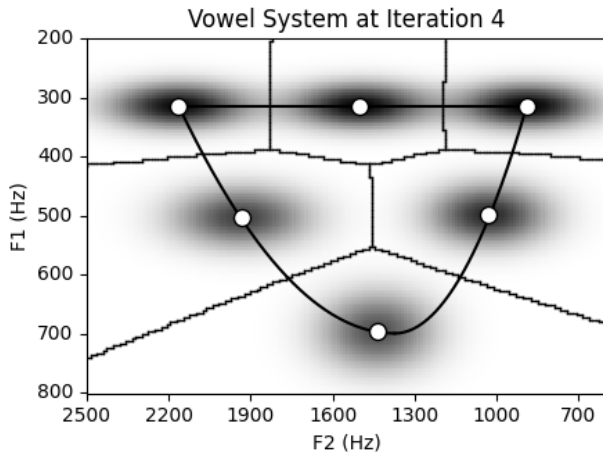
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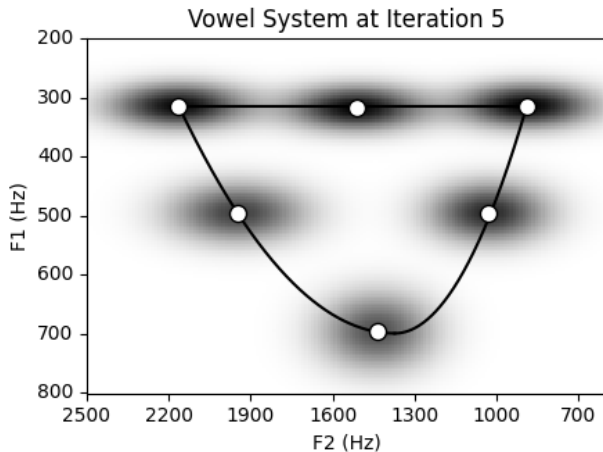
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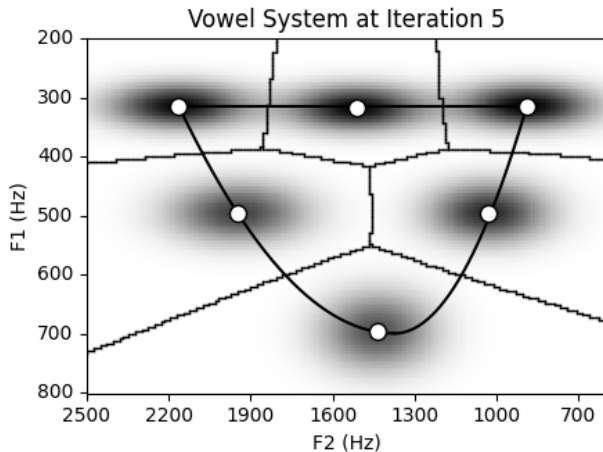
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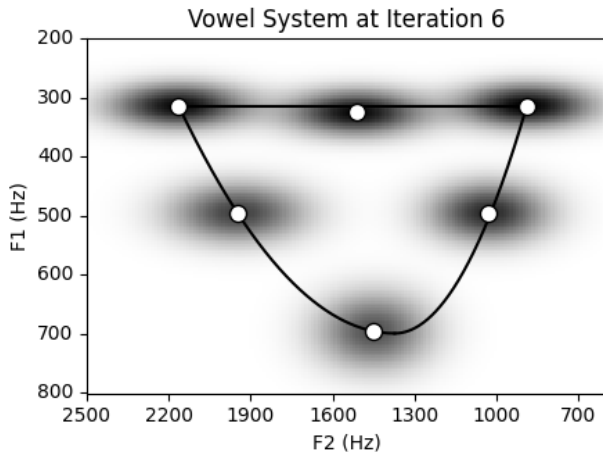
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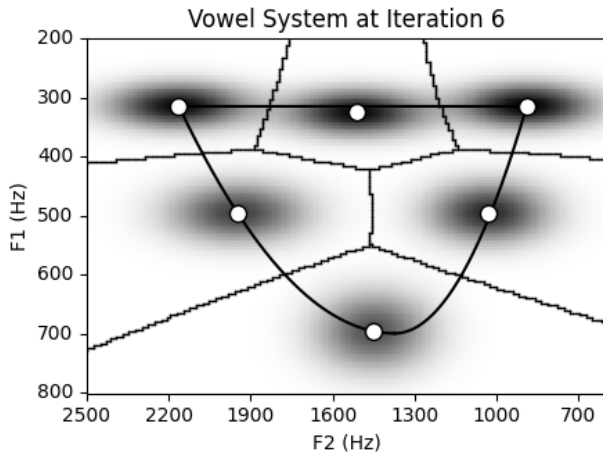
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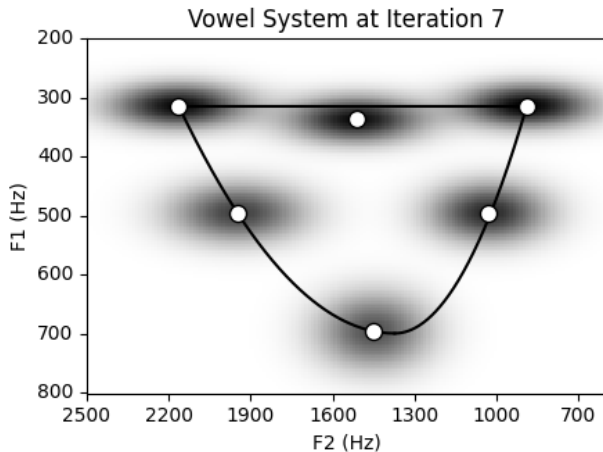
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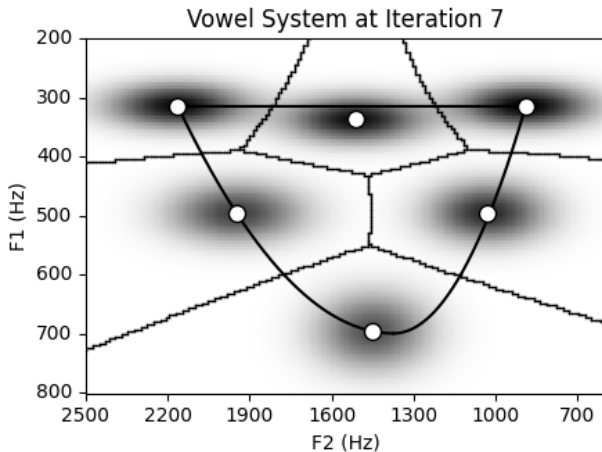
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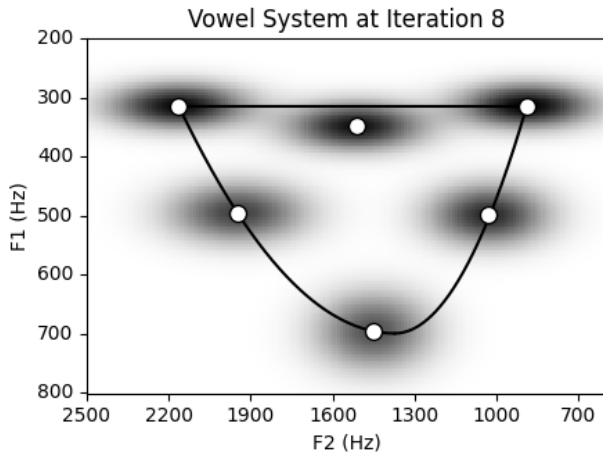
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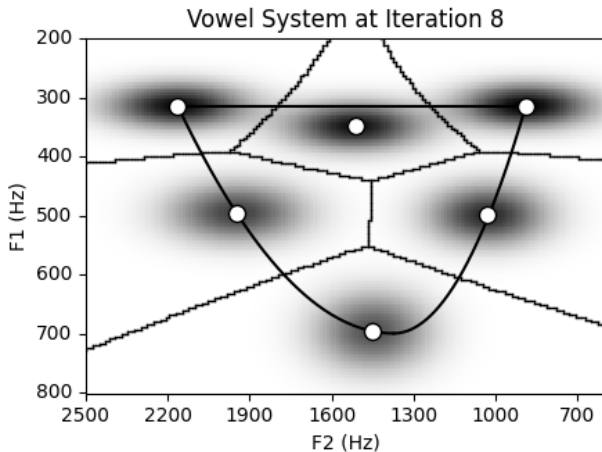
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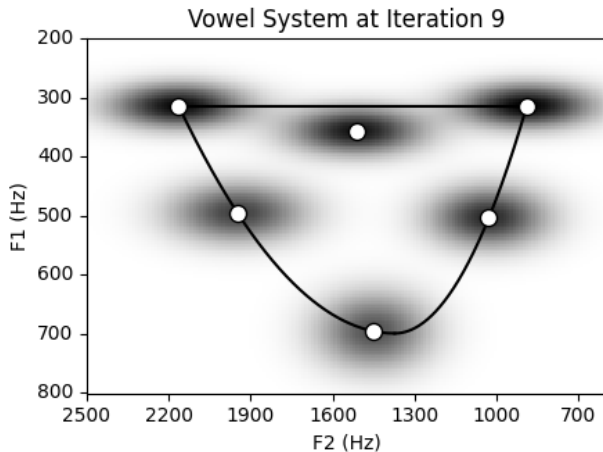
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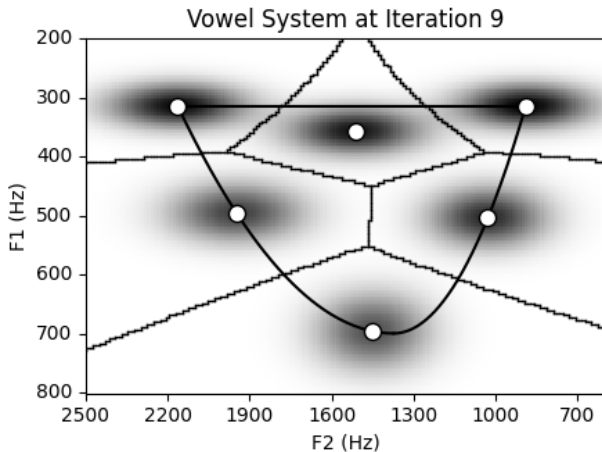
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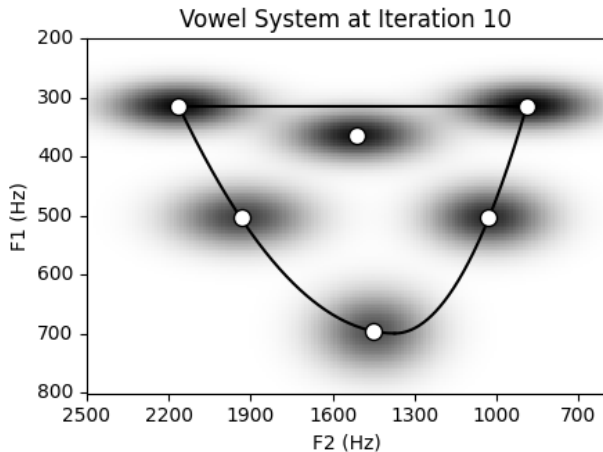
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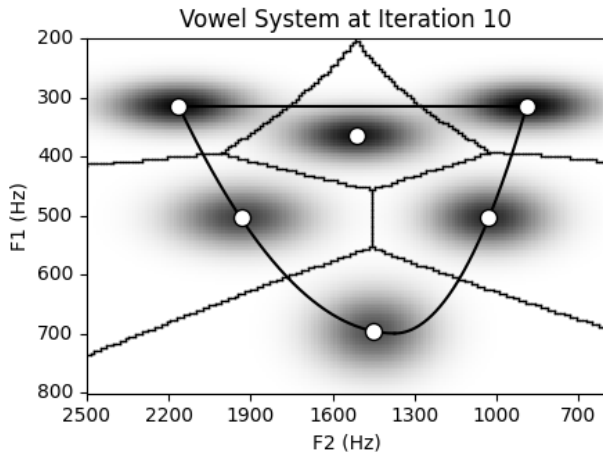
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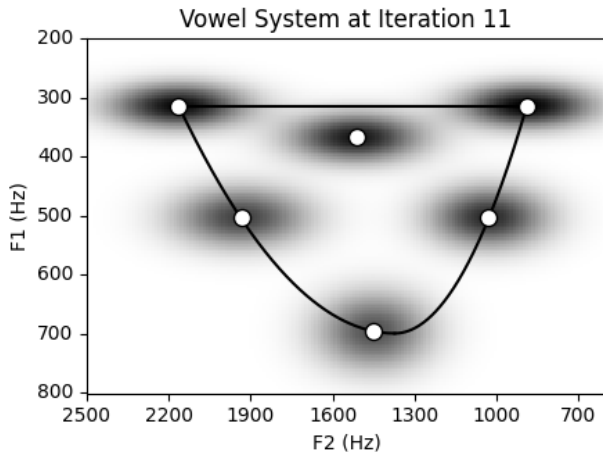
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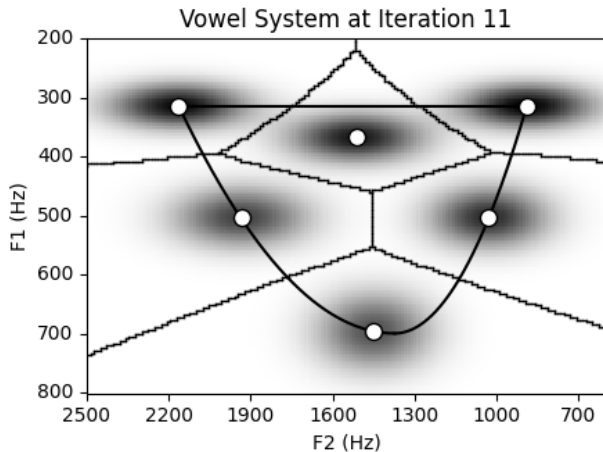
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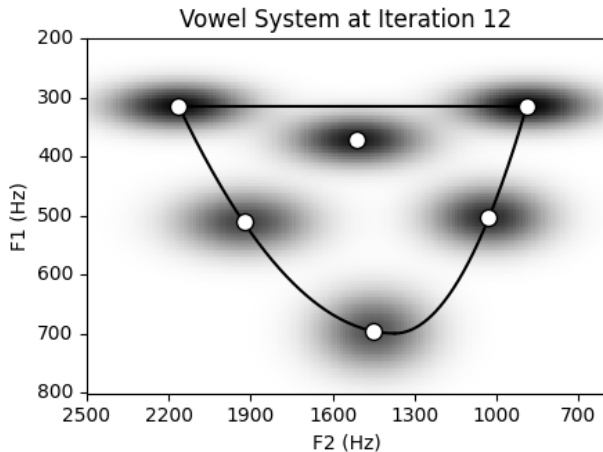
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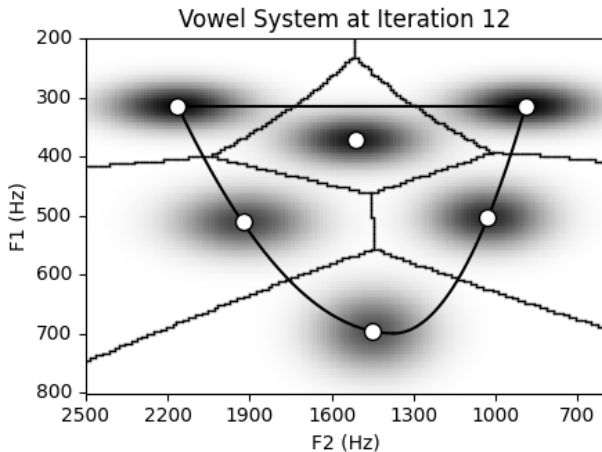
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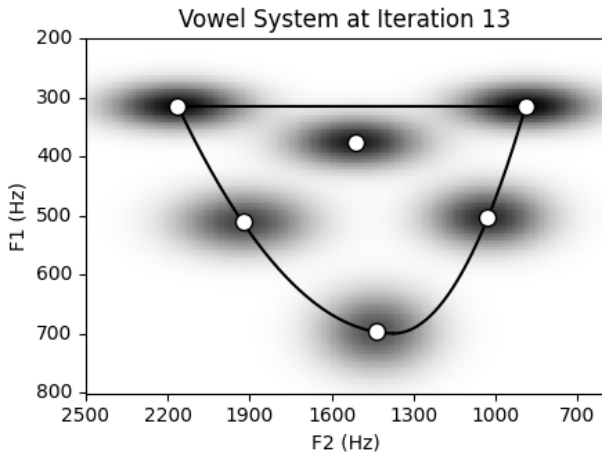
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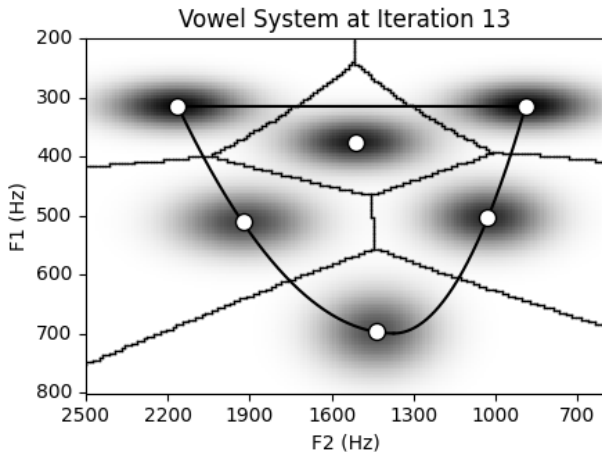


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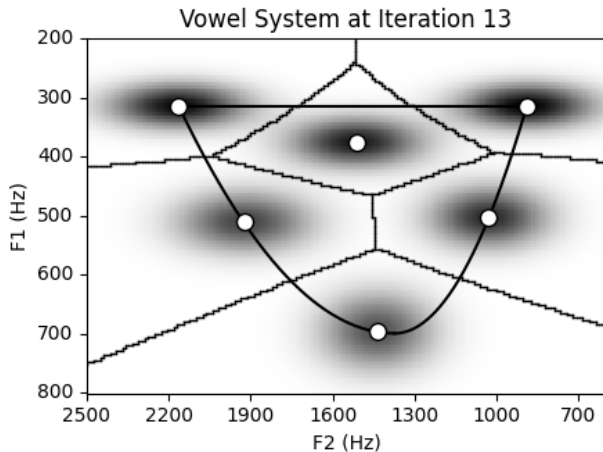


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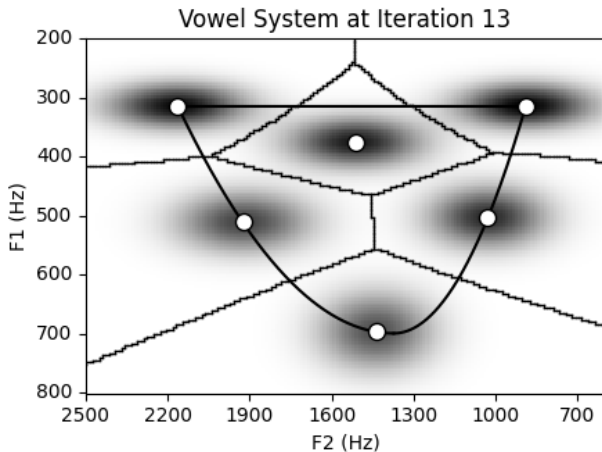
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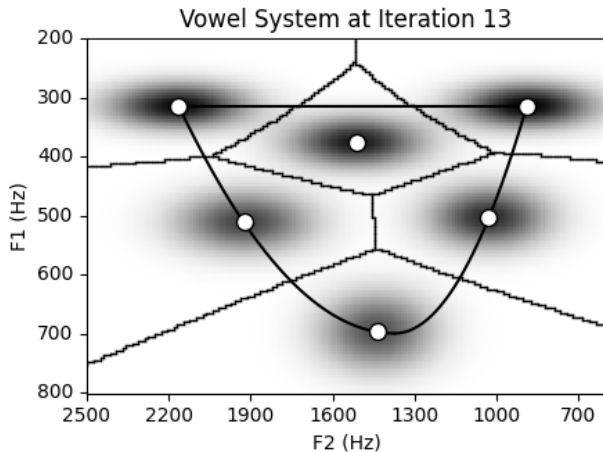
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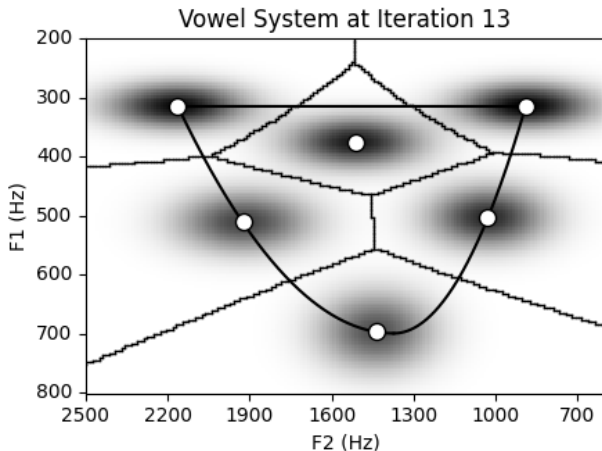
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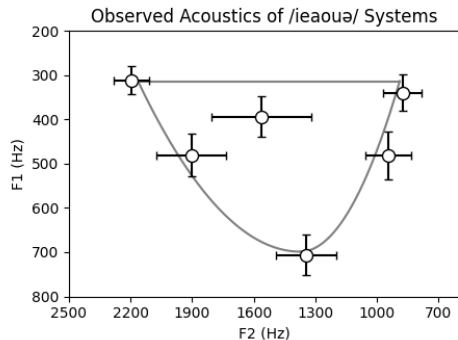
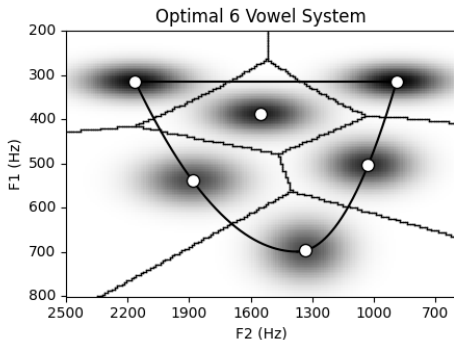
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- /ieɛaou/ is local optimum
- /ieaouə/ is global optimum



Acoustic Structure of /ieaouə/

Optimal 6V system matches acoustic structure [ieaouə] reported by Becker-Kristal (2010)



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 - e.g. /ieaouɨ/ vs. /ieaouə/ appears to be an artifact of transcriber choice in UPSID for a single system with interior vowel [ə] intermediate between [ɨ] and [ə]

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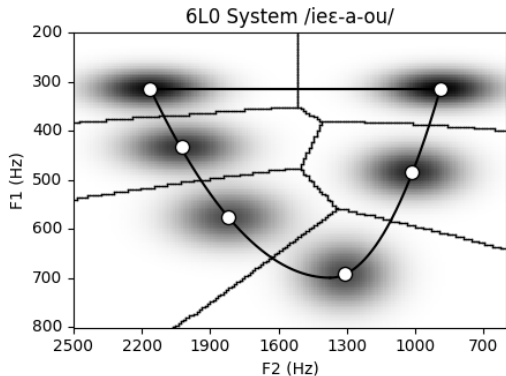
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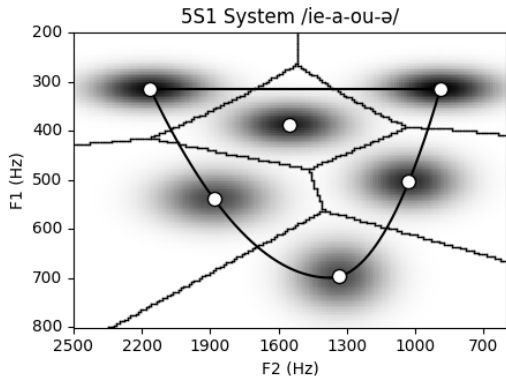
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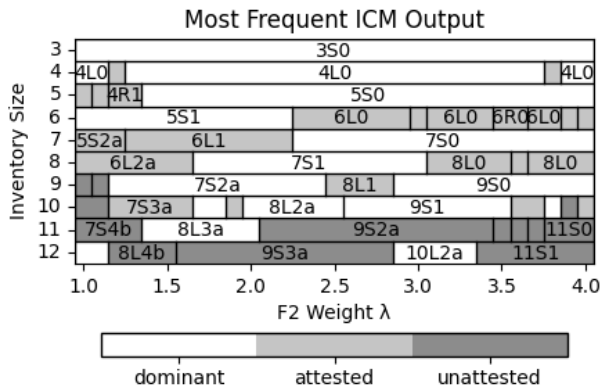


Typological Dominant Systems in the Baseline

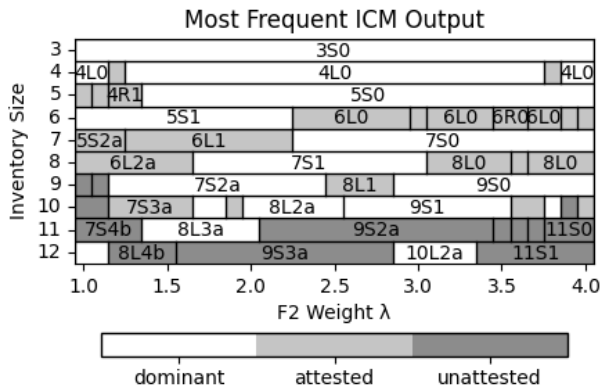
Most common system in UPSID by number of vowels:

Inventory Size	Dominant System	Example IPA	Frequency
3 (n=19)	3S0	/iaʊ/	1.000
4 (n=25)	4L0	/ieaʊ/	0.560
5 (n=109)	5S0	/ieaou/	0.890
6 (n=60)	5S1	/ieaouə/	0.683
7 (n=44)	7S0	/ieɛaou/	0.523
8 (n=19)	7S1	/ieɛaouə/	0.421
9 (n=24)	7S2a	/ieɛaouiə/	0.292
	9S0	/iieɛaouu/	0.292
10 (n=8)	8L2a	/iieɛaouiə/	0.250
	9S1	/iieɛaouuə/	0.250
11 (n=4)	8L3a	/ieɛæaouyøə/	0.667

Predicting Typologically Dominant Vowel Systems

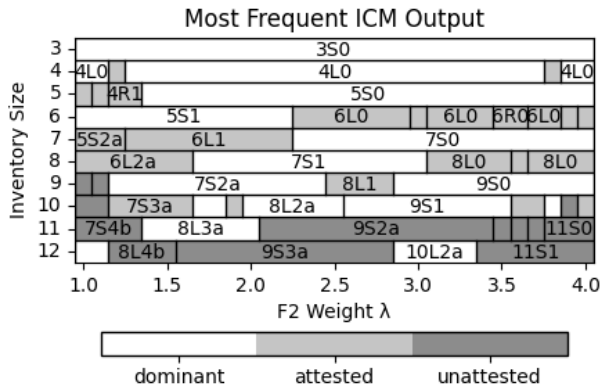


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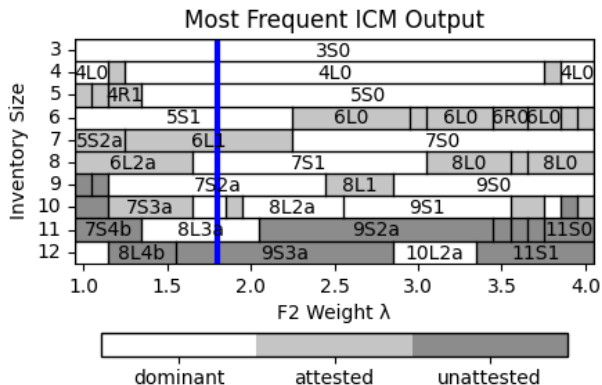
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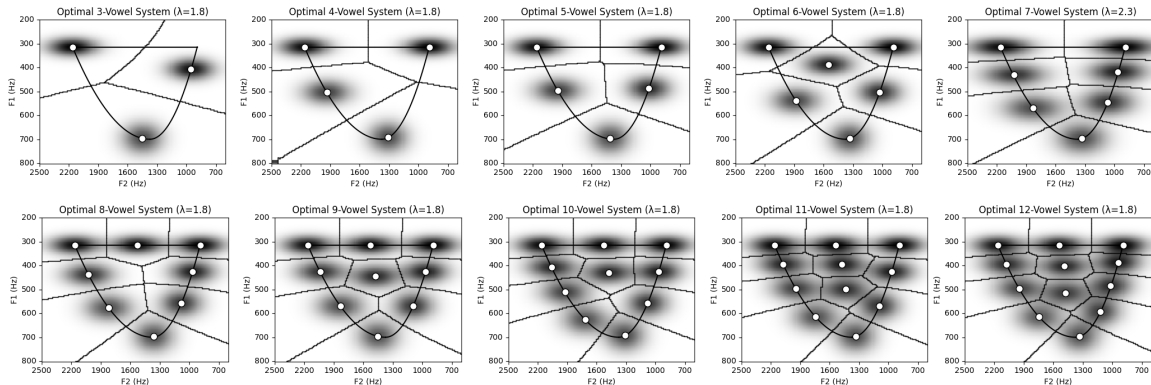


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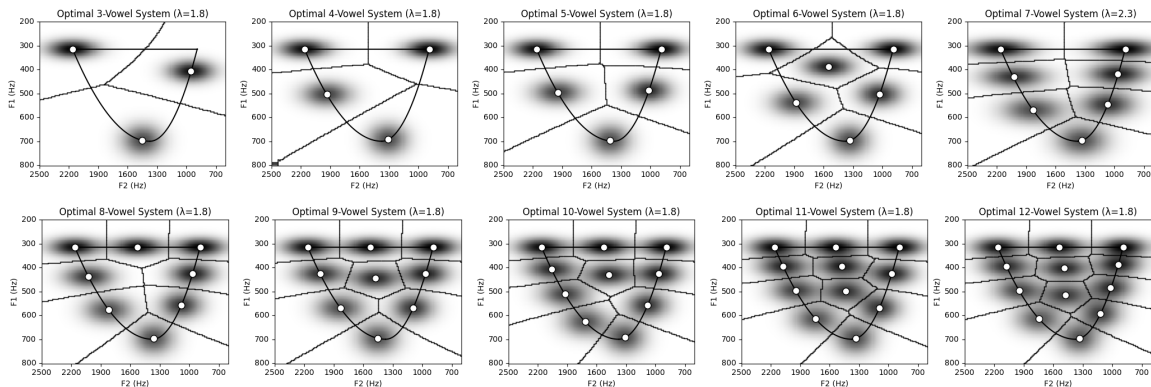
- Best model ($\lambda = 1.8$) predicts dominant system for 8/10 inventory sizes

Optimal Systems Visualized



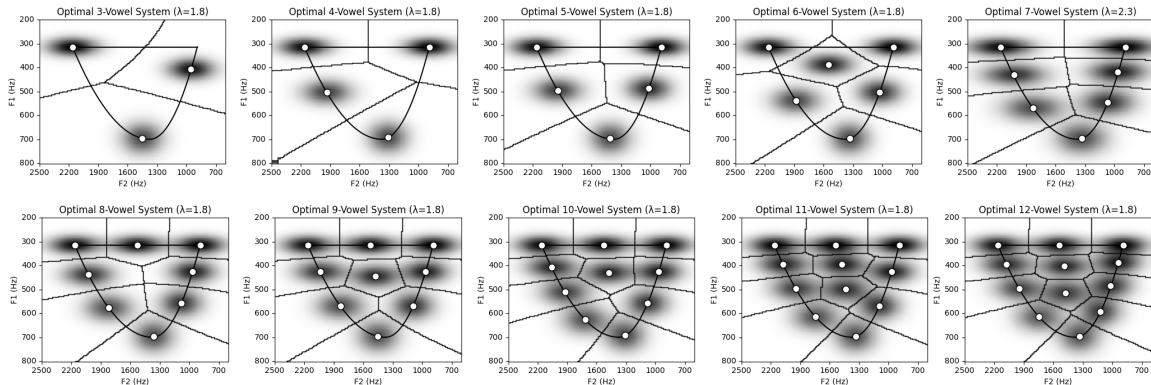
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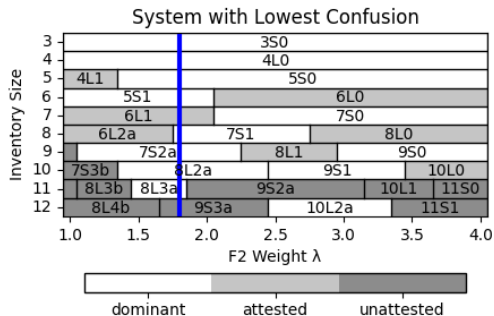
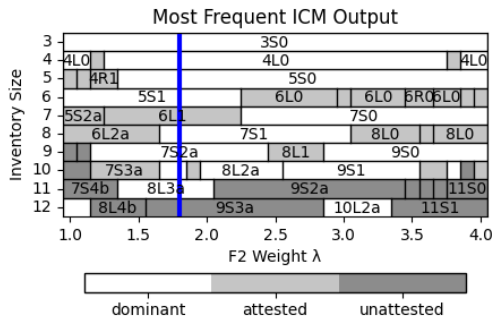
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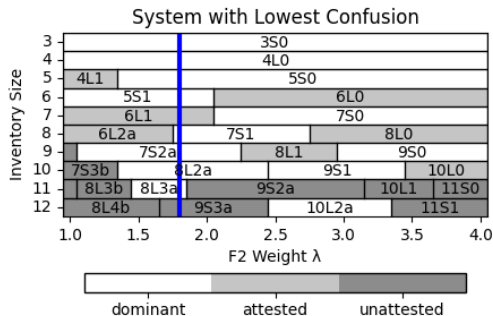
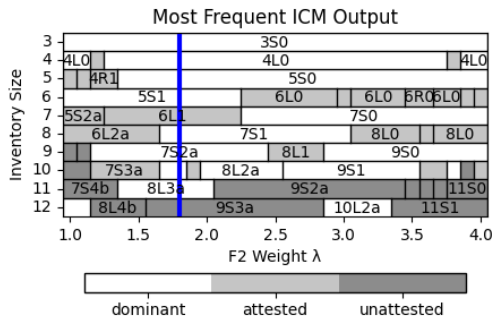
Improvement for larger inventories appears thanks to ability to predict interior vowels

Dominant Systems Optimize Confusion



Most frequent ICM outputs are both typologically dominant and perceptually optimal

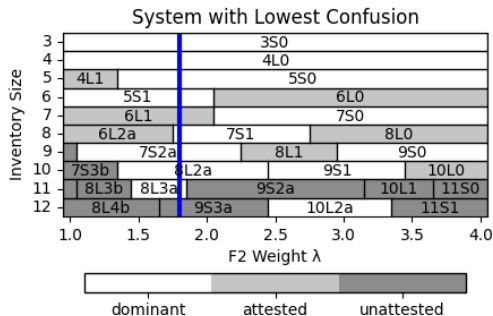
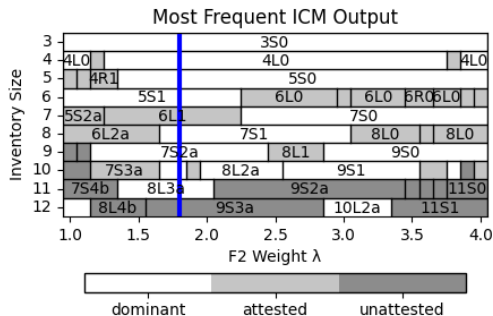
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Dominant Systems Optimize Confusion



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- **desideratum 3** ✓
- Optimal systems common because they emerge more easily through sound change

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Lower KL divergence = better model

Predicting Typological Frequencies

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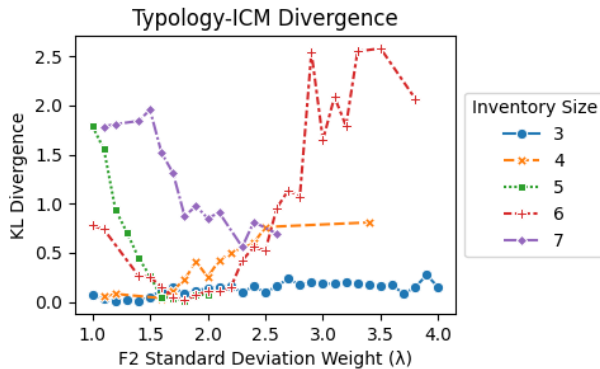
Kullback-Leibler (KL) divergence

$$D_{KL}(P||Q) = \sum_S P(S) \log \left(\frac{P(S)}{Q(S)} \right)$$

where:

- P = typological frequencies
- Q = model emergence rates

Lower KL divergence = better model



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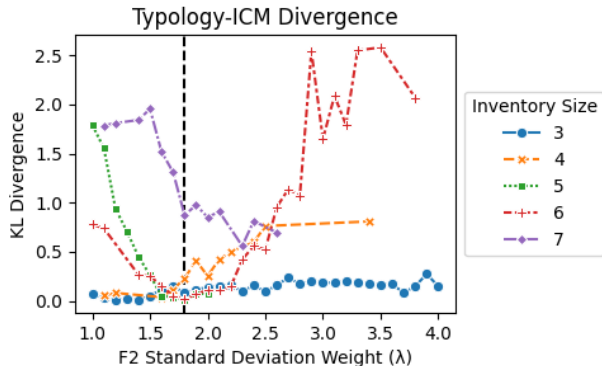
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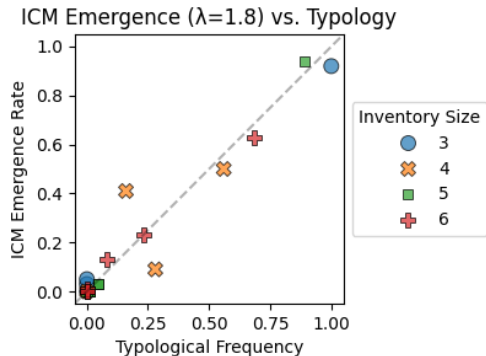
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Again, $\lambda = 1.8$ yields best overall model

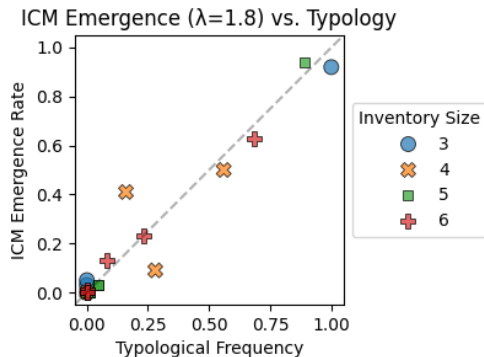


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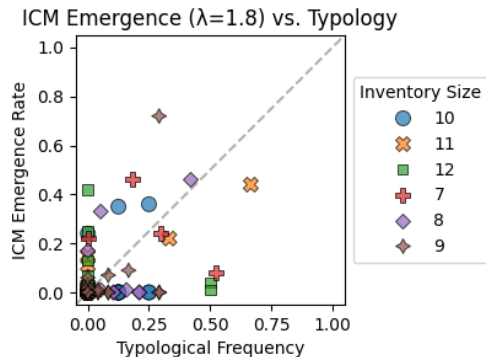
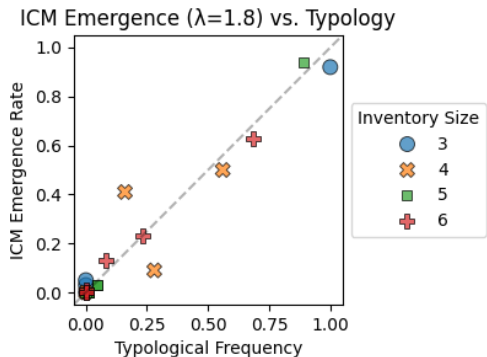


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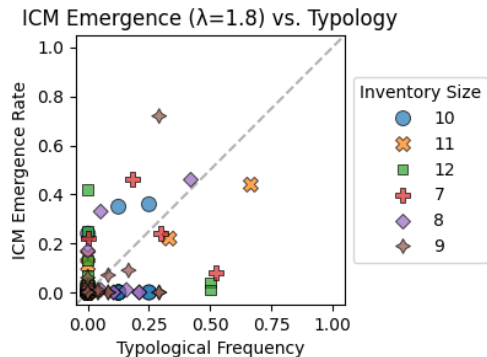
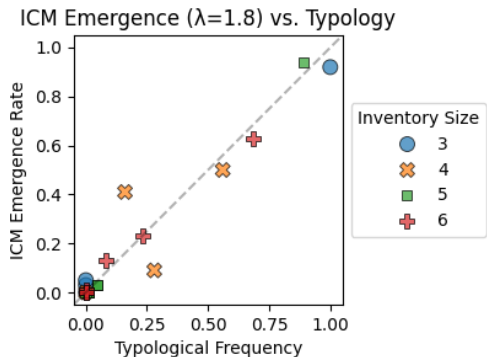
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Predicting Typological Frequencies



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Worse correlation from 7V: 2D space too idealized? Random initial systems problematic?

Predicting Typological Frequencies

Inventory Size	Kullback-Leibler Divergence	
	de Boer (1999)	ICM ($\lambda = 1.8$)
3	0.248	0.084
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- **desideratum 4** ✓

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 - with Coralie Cram & Megha Sundara

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Confusion minimization over larger sound sequences

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 - cf. functional load hypothesis (Gilliéron, 1918; Wedel et al., 2013)

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Thank You!

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Implementing ICM over the Vowel Space

boundaries on the vowel space

We will consider vowel distributions with F1 and F2 means (μ_1, μ_2) in Hz limited by the following equations, adapted from Liljencrants & Lindblom (1972) to match the space reported by Becker-Kristal (2010) for /ieaou/:

$$314 \leq \mu_1 \leq 699 \quad (1)$$

$$\mu_2 \leq 1373 + (2164 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}} \quad (2)$$

$$\mu_2 \geq 1373 + (889 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}} \quad (3)$$

N.B. – Simulations here use Hz (not psychoacoustic scales used in past models) since phonetic overlap does not depend on choice of scale.

Implementing ICM over the Vowel Space

set of possible vowel distributions

Vowels are modeled as bivariate normal distributions with 0 covariance:

$$(F1, F2) \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \right) \quad (4)$$

where:

- formant means (μ_1, μ_2) are limited by the previous equations, and
- standard deviations (σ_1, σ_2) are calculated as a function of formant means using a linear regression model fit to a crosslinguistic dataset.

Implementing ICM over the Vowel Space

formant standard deviation model

The standard deviation model was fit to z-scored data for 19 speakers of 21 languages from 10 studies^a with random effects for speaker and study:

$$z_1 = \frac{\mu_1 - 498.7212}{163.5568} \quad z_2 = \frac{\mu_2 - 1460.05}{543.4261} \quad (5)$$

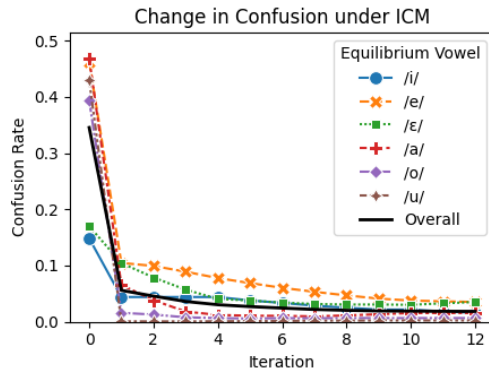
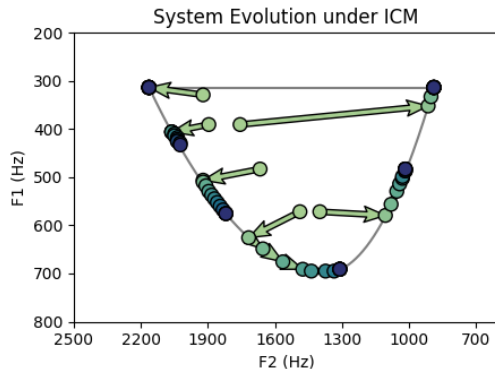
$$\sigma_1 = 34.9728 + 8.0738z_1 - 0.2633z_2 - 0.3906z_1z_2$$

$$\sigma_2 = \lambda(86.803 - 6.611z_1 + 6.741z_2 + 5.136z_1z_2)$$

^aHoltse, 1972; Thongkum, 1986; McCord, 1989; Svantesson, 1995; Olson, 2001; Calamai et al., 2003; Calamai, 2004; Thornell and Nagano-Madsen, 2004; Chiang and Chiang, 2005; Mayr and Davies, 2009

Following past dispersion models, this model also experiments with F2 scaling, implemented as a weight $1.0 \leq \lambda \leq 4.0$ on σ_2 .

Each ICM Iteration Reduces Overall Confusion



Idealizations of Maximum-Likelihood Behavior

Maximum-likelihood behavior makes two idealization:

- ① Prior probabilities are ignored
 - prior probabilities vary from language to language
 - More relevant for language-specific modeling than language-general typological models
 - also ignored in past Bayesian models (e.g. Clayards et al., 2008)
- ② Categorical perceptual boundaries are perfectly sharp
 - Human phonetic categorization follows more gradient sigmoid functions
 - Can be modeled if categorization tracks posterior probability (Clayards et al., 2008; Feldman et al., 2009; Kleinschmidt, 2019)
 - Simulations with posterior-tracking vs. maximum-likelihood behavior are qualitatively similar
 - Sharp boundaries give intuitive definition: listener confusion = phonetic overlap

Probabilistic Categorization

Actual human phonetic categorization is *not* perfectly categorical

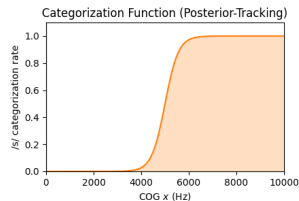
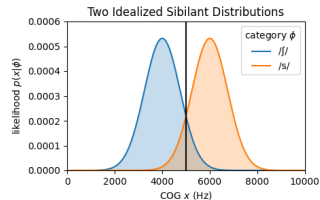
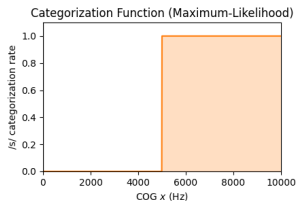
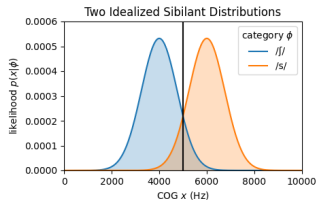
We can use posterior probabilities for probabilistic categorization

maximum-likelihood categorization

$$C(x) = \underset{\phi \in \mathcal{S}}{\operatorname{argmax}} p(x|\phi)$$

posterior-tracking categorization

$$P(C(x) = \phi) = \frac{p(x|\phi)P(\phi)}{\sum_{\phi' \in \mathcal{S}} P(\phi')}$$



Categorical (Maximum-Likelihood) vs. Gradient (Posterior-Tracking) ICM

More realistic probabilistic listener behavior does not significantly alter behavior of ICM

